# Subsidies, Cost Trends, and Pass-Through: Evidence from the Residential Solar Market

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**Abstract**

Solar subsidies are scheduled to fall as deployment grows, while installation costs change over the same period, so pass-through estimates that rely on subsidy timing can pick up cost trends. I study two Massachusetts programs that pay homeowners per kilowatt-hour, first through tradable certificates and then through a fixed rate. I identify pass-through from differences in the subsidy across roofs that the same installer equips in the same month, which arise because the payment depends on each roof's generation. I find that buyers receive about 85 percent of the subsidy under both designs, whereas a comparison over time suggests a 26-point gap. Installers raise prices only slightly for roofs that earn larger subsidies, which, with log-concave demand, bounds their markups at 2–7 percent of price.

# 1 Introduction

Governments subsidize residential solar heavily. The U.S. federal government alone spent $7.5 billion on solar subsidies in fiscal year 2022, almost all of it through tax credits (U.S. Energy Information Administration, 2023), and states and utilities add rebates, net metering, and payments for the electricity a system generates (Borenstein, 2017). Production-based payments are a common design: feed-in tariffs and tradable Solar Renewable Energy Certificates (SRECs) pay owners per kWh, and Massachusetts in 2018 and New Jersey in 2021 replaced market-priced certificates with administratively set payments. Whether any of these subsidies lowers the cost of adoption depends on pass-through, the share of the subsidy that reaches buyers rather than installers. Where installers capture part of it through higher prices, adoption falls and each unit of new capacity costs the government more.

Measuring pass-through is difficult because subsidies rarely change while everything else stays fixed. Existing estimates for solar identify pass-through from the timing of the subsidy within one state: step-downs in the California Solar Initiative rebate (Pless and van Benthem, 2019; Dong, Wiser and Rai, 2018), step-downs in Connecticut's rebate (Gillingham and Tsvetanov, 2019), and a change in the federal investment tax credit (Podolefsky, 2013). Solar subsidy schedules, however, are designed to decline as deployment grows. California's rebate stepped down as cumulative capacity reached preset targets, and Massachusetts's fixed rate falls as capacity blocks fill. The subsidy therefore falls when installation costs fall, and a regression of prices on the subsidy mixes how the subsidy is shared with how costs and the subsidy move together. Estimates of this kind depend on how the path of costs is modeled: in Pless and van Benthem (2019), host-owned pass-through ranges from 0.70 to 1.01 across their specifications of the time trend. Studies of tax incidence in other markets address common cost shocks with a comparison group that the tax does not reach (Kosonen, 2015; Kopczuk et al., 2016; Butters, Sacks and Seo, 2022), but a solar subsidy reaches every installation in a utility territory at a given time.

This paper builds the comparison group inside the treated market and month. I study

two production-based programs in Massachusetts. Under the Solar Carve-Out II program (2014–2018), owners earned SRECs sold at market prices. Under its successor, the Solar Massachusetts Renewable Target (SMART) program, they receive a fixed rate per kWh that is set when the application is approved and declines as capacity blocks fill. Because both programs pay per kWh, a project's subsidy depends on how much electricity its roof will generate. I merge regulatory data from the Massachusetts Department of Energy Resources (DOER) with Lawrence Berkeley National Laboratory's Tracking the Sun database for more than 39,000 host-owned residential systems installed between 2014 and 2023, geocode each project's street address, and model each system's annual generation with the National Renewable Energy Laboratory's PVWatts calculator from its location, size, tilt, azimuth, and equipment. About 91% of the variance in modeled capacity factors is within towns. Roofs that the same installer equips in the same month therefore earn different subsidies, and this variation does not depend on the policy calendar. My preferred specification compares roofs within installer-by-year, town-by-year, and month cells, and comparing roofs of the same installer in the same month gives the same estimates. An upfront rebate per watt is nearly the same for every roof, so this design is available because the subsidy is paid for production.

The first finding is that the two designs pass through the subsidy at the same rate. Comparing roofs, pass-through is 0.86 under Solar Carve-Out and 0.85 under SMART. The difference is −0.02 with a standard error of 0.085, so the 95% confidence interval runs from −0.18 to 0.15. The subsidy in both programs is built from the same model of generation, so error in modeled generation does not by itself create or hide a difference between them. It does pull both levels toward one, so true pass-through may be somewhat lower. Installers' own production estimates, which SMART applications report, agree only partly with modeled generation within these cells, and using them as an alternative measure gives SMART pass-through of 0.74–0.77. Those estimates are prepared for sales and exist only for SMART, so I treat them as a sensitivity check. The programs differ in the form of the payment, in how predictable it is, and in size: the average Solar Carve-Out subsidy is about 3.4 times the

average SMART subsidy per kW ($1,923 versus $566 in present value), about 45% and 15% of the price. Buyers nonetheless capture about 85 cents of each dollar under both. To my knowledge, these are the first estimates of the pass-through of a production-based subsidy and the first comparison of a certificate-based and a fixed-rate design within one market.[1]

The second finding is that estimates that rely on the timing of the subsidy follow the cost trend. Under SMART, the subsidy kept falling after 2020 while prices rose, which biases a timing-based estimate up. A naive comparison with year fixed effects gives 1.02, and a stacked event study around the step-downs in SMART rates, a design built to avoid trends, gives 1.27, against 0.85 across roofs. Measurement error cannot produce this gap, because it pulls estimates toward one rather than above it, and pass-through above one is inconsistent with the log-concave demand I estimate unless marginal cost falls with volume. Under Solar Carve-Out, the subsidy and installation prices fell together, which biases a timing-based estimate down. The naive estimate, 0.76, lies below the cross-roof estimate of 0.86, as predicted, but error in modeled generation also pulls the cross-roof estimate toward one, and the estimate identified from roof orientation alone is 0.76. The data therefore cannot show that the naive Solar Carve-Out estimate is biased, but the timing-based estimates move with the cost trend in their window: a regression discontinuity at the January 2017 cut in the SREC factor falls from 1.03 to 0.71 as the window widens from 90 to 365 days. The naive gap of 0.26 between the programs therefore comes mostly from the SMART estimate, and year-month fixed effects alone remove most of it.

These results also show how much market power installers have. Under a uniform price, a subsidy specific to one roof would be passed through in full, so the shortfall of cross-roof pass-through below one measures how far installers price the roof. Installers quote each job individually, and prices differ by about $450–490 per kW across buyers of the same installer in the same month, yet a roof that earns $59–137 more modeled subsidy per kW is charged only

[1] Comparisons in other settings find that the form of an incentive can affect pass-through: car buyers capture 70 to 90 percent of a customer rebate but 30 to 40 percent of a dealer discount (Busse, Silva-Risso and Zettelmeyer, 2006), and the point at which a diesel tax is collected changes its pass-through (Kopczuk et al., 2016).

$6–20 more. With log-concave demand under SMART, the sufficient-statistics framework of Weyl and Fabinger (2013) turns this into a bound: installer markups are at most 2–7% of price, and they exceed 10% only if more than about half of the remaining variation in modeled generation is error. The bound rests on demand curvature that is estimated with limited precision. The other channels through which installers could earn markups, such as local concentration, the number of competitors, and buyer income, leave little trace in pass-through or prices, although SMART installers price the roof somewhat more in months of high statewide demand. Pass-through alone cannot rule out a markup common to all buyers in a market.

This paper contributes to three literatures. The first studies the incidence of taxes and subsidies and the variation that identifies it. Reduced-form studies identify incidence from dated policy changes, from rebate step-downs and tax credit changes (Sallee, 2011; Busse, Silva-Risso and Zettelmeyer, 2006) to tax changes compared with untaxed goods or regions (Kosonen, 2015; Kopczuk et al., 2016; Butters, Sacks and Seo, 2022), from national credit prices (Lade and Bushnell, 2019), and from before–after comparisons (Kenkel, 2005). The variation also determines the estimand: in U.S. oil refining, pass-through ranges from near zero for a cost shock specific to one firm to about one for a shock common to the industry (Muehlegger and Sweeney, 2022), and California's electric vehicle rebates, which differ by income, identify the price recipients pay relative to non-recipients (Muehlegger, Rapson et al., 2018).[2] I show that in a market where the subsidy schedule tracks costs and no untreated group exists, comparing buyers who face different subsidies at the same time recovers pass-through, and that timing-based designs, including a regression discontinuity and an event study, move with the cost trend in their window. Where prices rose as the subsidy fell, they give pass-through above one.

[2]Related work studies hybrid vehicle subsidies with endogenous quality and costly bargaining (Gulati, McAusland and Sallee, 2017), electric vehicle subsidies across countries (Barwick et al., 2023), the incidence of agricultural subsidies on farmland rents (Kirwan, 2009), and the welfare effect of a consumer subsidy restricted by price ceilings (Fan and Zhang, 2022). Incomplete pass-through can also reflect local costs and markup adjustment (Nakamura and Zerom, 2010), and firms that set prices nationally may respond differently to local cost shocks (Butters, Sacks and Seo, 2022).

The second literature estimates the pass-through of solar subsidies and studies how subsidy design affects buyers. Most of this evidence comes from upfront rebates, and it finds that buyers capture most but not all of the subsidy: 78% for host-owned systems in California (Pless and van Benthem, 2019), 84% in Connecticut (Gillingham and Tsvetanov, 2019), and close to complete pass-through in other work on California (Dong, Wiser and Rai, 2018). Pless and van Benthem (2019) find more-than-complete pass-through for third-party-owned systems, and for the federal investment tax credit, reported prices also reflect the incentive to overstate system costs (Podolefsky, 2013). Solar adoption responds strongly to rebates (Hughes and Podolefsky, 2015; Crago and Chernyakhovskiy, 2017), buyers discount production-based payments heavily (De Groote and Verboven, 2019), and upfront subsidies induce more adoption while production subsidies are more efficient (Burr, 2016).[3] I provide estimates for production-based subsidies paid through certificates and through fixed rates, and I find that the two designs pass through the same share once the timing of the subsidy is absorbed.

The third literature uses pass-through to learn about market power. Pass-through tends to rise with competition, from 13% in the least competitive Medicare Advantage markets to 74% in the most competitive (Cabral, Geruso and Mahoney, 2018) and, for fuel taxes on Greek islands, from 0.4 in monopoly markets to 1 in markets with four or more competitors (Genakos and Pagliero, 2022). It has been used to measure incidence under imperfect competition (Ganapati, Shapiro and Walker, 2020) and, with convex demand, to detect market power from more-than-complete pass-through (Pless and van Benthem, 2019), while high pass-through can also arise from cost shocks common to all firms (Fabra and Reguant, 2014). I bound markups with roof-specific pass-through, which removes the supply side from the pass-through formula, together with estimated log-concave demand.

[3]Other studies compare the effectiveness of residential solar incentives (Matisoff and Johnson, 2017), estimate how residential solar demand depends on net metering and electricity prices (Ros and Sai, 2023), and characterize optimal subsidies for residential solar (Colas and Reynier, 2026). Consumer subsidies can also lower production costs through induced innovation (Gerarden, 2023), and the phase-out of a subsidy shapes the dynamics of demand (Hu, Yin and Zhao, 2023).

Section 2 describes the two programs, Section 3 the data and the construction of the subsidy, and Section 4 the empirical strategy and why timing variation is confounded. Section 5 compares pass-through across the two programs, Section 6 shows that the estimates that rely on timing follow the cost trends, and Section 7 addresses threats to the cross-roof design. Section 8 draws the implications for market power, and Section 9 concludes.

# 2 Institutional Background

When a household in Massachusetts installs a solar system, one of two production-based subsidies is available, depending on the policy regime at the time of installation. If a qualifying system entered commercial operation between January 2008 and November 2018, the owner was eligible to receive certificates under the Solar Carve-Out program, which launched in 2010 and covered systems operating since 2008. If it entered commercial operation after November 2018, the owner is entitled to an incentive payment through the electricity bill, on top of net metering, under the SMART program. This section describes the two programs, and Section 3 describes the data and the method I use to compute the subsidy.

## 2.1 Solar Carve-Out

The Solar Carve-Out is part of Massachusetts's Renewable Portfolio Standard (RPS) and requires that a certain percentage of electricity come from solar resources. The program launched on January 1, 2010, and its requirement increased over time until it reached a capacity target of 400 MW. The state expanded the program in April 2014 with Solar Carve-Out II, which set a total goal of 1,600 MW of solar PV capacity.

Energy suppliers, including utilities and competitive energy suppliers, are required to purchase Solar Renewable Energy Certificates (SRECs) to demonstrate compliance or make an alternative compliance payment (ACP). The NEPOOL Generation Information System (NEPOOL GIS) tracks solar generation and issues SRECs quarterly to solar generation unit

owners at a rate of 0.45–1.0 SREC per MWh, depending on system characteristics (the SREC factor). For example, a system smaller than 25 kW authorized to interconnect before January 8, 2017, receives a factor of 1.0. Each qualified solar generation unit is entitled to receive SRECs for 40 calendar quarters as long as the unit remains operational. After 40 calendar quarters, the system earns Renewable Energy Certificates (RECs) instead, which are much less valuable than SRECs.[4]

Solar generation unit owners can either market and trade SRECs independently or join an aggregator that harvests and trades SRECs on their behalf. Trading platforms and aggregators typically charge fees of 1–10% of the proceeds depending on system size, with small residential systems (under 50 kW) commonly paying 10%. The Massachusetts Department of Energy Resources (DOER) holds an annual fixed-price clearinghouse auction from May 16 to June 15 for SRECs not sold during the previous year. The auction does not guarantee a sale: certificates that remain unsold through it can still be traded in the market for the next two compliance years. The ACP thus caps SREC prices, but the auction price is not a floor. In the Solar Carve-Out's launch year, the ACP was set at \$600/MWh, while the net clearinghouse auction price was \$285/MWh (a \$300 price less the 5% auction fee).[5] For Solar Carve-Out II, the ACP and auction price schedules were published at launch, whereas under the original program the ACP was adjusted over time. Publishing the schedules narrowed the range of likely SREC prices but did not remove price risk: as Figure 1 shows, market prices sat below the auction price for several years.

Figure 1 shows historical market prices for SREC II certificates alongside the ACP and clearinghouse auction prices. In the first few years, market prices were consistently below auction prices because of oversupply. In later years, market prices for SRECs followed the

[4]The vintage of a certificate is the calendar year in which the electricity is generated. For instance, electricity generated during 2014 earns MA2014(II) certificates, where "MA" denotes Massachusetts, "2014" the generation year, and "II" the program phase, Solar Carve-Out II. Certificates of different vintages form distinct market segments and can trade at different prices, depending on factors such as program caps, ACP levels, and the balance between supply and demand within each vintage year.

[5]Under the original program, market prices for SRECs tracked closer to the ACP, and returns exceeded twice the retail electricity price in 2010.

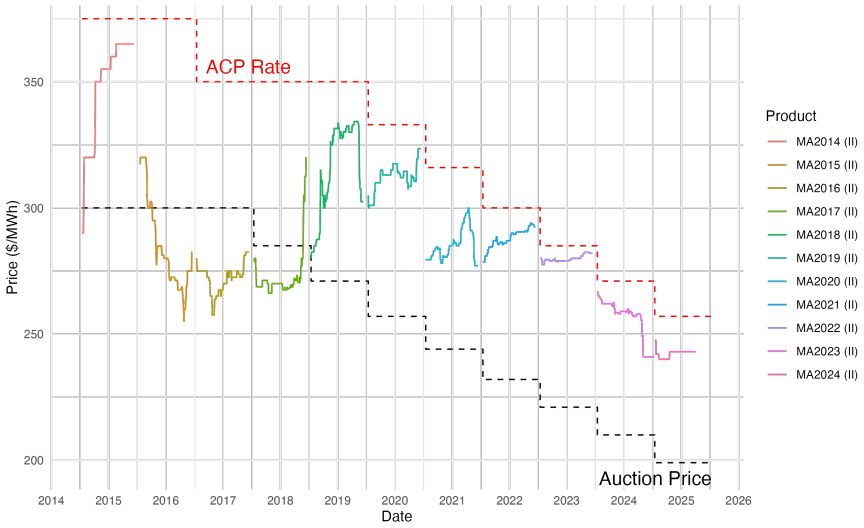



Figure 1: SREC II Market Prices with ACP and Auction Prices. The auction price is plotted gross of the 5% auction fee. An owner selling through the auction receives 95% of the plotted value ($285/MWh in 2014–2016). Each vintage is shown over its trading window, from July 15 of the vintage year to July 14 of the following year.

declining trend of the ACP. The net clearinghouse price remained constant at $285/MWh from 2014 to 2016 and then fell by 5% per year.[6]

## 2.2 SMART

DOER created the Solar Massachusetts Renewable Target (SMART) program to further encourage solar development in Massachusetts after the Solar Carve-Out II target was met. The program launched on November 26, 2018. Under SMART, the three investor-owned utilities (IOUs) in Massachusetts (Eversource, National Grid, and Unitil) compensate solar system owners for the electricity their systems generate, in exchange for the RECs attached to this generation.

The actual incentive rate for a given system is determined by four factors: the base compensation rate, household income status, whether the system includes a storage component, and the value of energy. The base compensation rate starts high and declines on a predetermined schedule as capacity blocks fill, and both block sizes and rate schedules vary by utility.

[6]The net clearinghouse price stood at $189.05/MWh in 2024.

The value of energy is set from three-year average retail electricity prices (including delivery and transmission charges) at the time of application. The SMART incentive rate equals the base compensation rate minus the value of energy. Small residential solar generation unit owners receive payments through the utility equal to this rate multiplied by their monthly generation, in addition to the bill savings their behind-the-meter systems already provide, and they can cash out a negative balance monthly. As electricity prices rise and capacity blocks advance, the SMART incentive rate decreases until it reaches zero.[7]

Table 1 shows the base compensation rates for generation units of 25 kW AC or less, by utility.[8]

Table 1: SMART Base Compensation Rates for Generation Units of 25 kW AC or Less

| Utility | Low income | Block 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| Fitchburg | Yes | \$0.358 | \$0.326 | \$0.298 | \$0.272 | — | — | — | — |
| | No | \$0.311 | \$0.284 | \$0.259 | \$0.236 | — | — | — | — |
| Mass Electric | Yes | \$0.358 | \$0.344 | \$0.330 | \$0.317 | \$0.304 | \$0.292 | \$0.280 | \$0.269 |
| | No | \$0.311 | \$0.299 | \$0.287 | \$0.275 | \$0.264 | \$0.254 | \$0.244 | \$0.234 |
| Nantucket | Yes | \$0.391 | \$0.328 | — | — | — | — | — | — |
| | No | \$0.340 | \$0.286 | — | — | — | — | — | — |
| Eversource E. | Yes | \$0.391 | \$0.375 | \$0.360 | \$0.346 | \$0.332 | \$0.319 | \$0.306 | \$0.294 |
| | No | \$0.340 | \$0.326 | \$0.313 | \$0.301 | \$0.289 | \$0.277 | \$0.266 | \$0.255 |
| Eversource W. | Yes | \$0.329 | \$0.315 | \$0.303 | \$0.291 | \$0.279 | \$0.268 | \$0.257 | \$0.247 |
| | No | \$0.286 | \$0.274 | \$0.263 | \$0.253 | \$0.243 | \$0.233 | \$0.224 | \$0.215 |

Base compensation rates in dollars per kWh by utility and capacity block. "Low income" indicates the rates for low-income customers.

SMART was most favorable early in its life cycle, when the incentive rate was considerably higher than the price of RECs. In my sample, the SMART incentive rate is at most \$0.20/kWh, with a median of about \$0.066/kWh, while RECs traded at around \$0.04/kWh.[9] SMART also offers more predictable cash flows than the Solar Carve-Out program, because

[7]System owners who do not wish to enroll in SMART can instead participate directly in the REC market, and DOER advises them to do so when the SMART rate becomes low. I do not observe how many owners opt out, because owners who do so do not appear in the SMART regulatory data.

[8]There are eight blocks for the three large utilities but only two and four for Nantucket and Fitchburg, respectively. Fitchburg is Unitil's operating company, and Mass Electric and Nantucket Electric are National Grid's operating companies in Massachusetts.

[9]Source: SRECTrade trading platform.

the rate is fixed and does not fluctuate with market conditions. The subsidy can therefore be computed precisely from modeled generation. Another key difference between SMART and Solar Carve-Out is that SMART requires system interconnection to one of the three IOUs. As a result, households served by municipal utilities are ineligible for the program.

## 2.3 Comparison of the Two Programs

Table 2 summarizes the two programs. Four differences could affect how subsidy value is divided between buyers and installers. First, the form of payment: an owner under Solar Carve-Out sells certificates whose price is set in a market capped by the ACP, whereas a SMART owner receives a fixed per-kWh rate. Second, the timing of information: the SMART rate is fixed when the application is approved, so the buyer and installer know the subsidy per kWh for ten years, while the SREC price the owner will eventually receive is unknown at the time of purchase. Third, the size of the subsidy: the mean present value is about $1,923 per kW under Solar Carve-Out and $566 per kW under SMART, roughly 45% and 15% of the average gross price, with almost no overlap between the two distributions. Fourth, coverage: Solar Carve-Out is open to all utility territories, while SMART is limited to customers of the three IOUs.

These differences suggest that pass-through could differ between the programs. A larger subsidy that arrives through a volatile market price is harder to price into a quote and may be more likely to strain installers' capacity, which would lower pass-through. A small, certain subsidy is easy to incorporate into a price and is less likely to move installer capacity, which would raise pass-through. Because the programs also differ in years, technology costs, and eligible households, a comparison of the programs over time cannot test these predictions. Section 5.2 tests whether pass-through differs once the timing of the subsidy is absorbed.

Two rules within the programs change the subsidy discretely, and Section 6 uses both. On January 8, 2017, the SREC factor for new small systems fell from 1.0 to 0.8, so a system that generates the same electricity receives 20% fewer certificates. Under SMART, the base

Table 2: Comparison of the Solar Carve-Out II and SMART Programs

| | Solar Carve-Out II | SMART |
|---|---|---|
| Period in sample | Jan. 2014–Nov. 2018 | Mar. 2019–Sep. 2023 |
| Form of payment | Tradable SRECs | Fixed rate per kWh |
| Rate | Market price, capped by the ACP | Fixed at approval |
| Payment term | 40 calendar quarters | 10 years |
| Payer | Retail electricity suppliers | The three IOUs |
| Eligibility | All utility territories | IOU customers only |
| Mean subsidy (NPV, 8%) | $1,923 per kW | $566 per kW |
| Share of gross price | ~45% | ~15% |

Subsidy is the present value per kW of DC capacity at an 8% discount rate with 0.5% annual degradation, from the analysis samples. Shares are ratios of the sample mean subsidy to the sample mean gross price per kW (systems of 25 kW or less).

rate of a utility steps down when one of its capacity blocks fills (Table 1).

# 3 Data

## 3.1 Data Sources and Sample

I obtain the solar installation data from DOER. For SMART projects, the DOER data report compensation rates as well as project characteristics, including location, utility, size, installation cost, installer, and technical details. The technical details, which include azimuth, tilt, and inverter rating, are essential for modeling generation accurately. The compensation rate reported in the data is the actual SMART incentive rate that system owners receive, which reflects any adders and the value of energy.

For Solar Carve-Out II projects, DOER reports similar variables but without technical details. To model annual generation as accurately as for SMART, I merge the DOER data with the Tracking the Sun database published by Lawrence Berkeley National Laboratory (LBNL).[10] All projects in my data appear in Tracking the Sun, along with their technical details. I match projects across the two sources on operation date, size, and total cost, which gives an almost one-to-one match (Appendix A). SREC prices are not available in the project data. I construct an SREC price series from a major SREC trading platform, SRECTrade, and from the price support regulations published on DOER's website.

The SREC sample spans January 1, 2014, to November 26, 2018, the period when Solar Carve-Out II was open to applications. The SMART sample begins on March 25, 2019, marking the debut of the incentive program, and ends on September 20, 2023, by which time most projects received zero compensation and the number of applications had dropped sharply. For SMART projects, I date each project by its Commercial Operation Date (Appendix A).

The sample includes only host-owned, small-scale (≤25 kW DC) residential projects without battery storage, for three reasons. First, the general net metering program has a capacity cap, so projects installed after the cap was reached might not be eligible for net metering.

[10] Tracking the Sun is a cleaned dataset that covers more than 60% of solar projects in the United States. LBNL cleans and consolidates information on solar projects from state energy agencies annually. DOER and the Massachusetts Clean Energy Center are both major data providers for LBNL.

Small systems are exempt from the cap, so restricting the sample to them ensures that every system is on net metering. Second, only host-owned projects have directly observed prices. For third-party-owned projects, the price reported to DOER is reportedly a "market" value rather than the true cost to system owners (Pless and van Benthem, 2019). Third, installations with battery storage qualify for a storage adder, which raises the incentive, and the storage components would distort the measured cost of the solar system. I also remove observations with a per-kW price above the 99th percentile or below the 1st percentile to exclude outliers.[11]

## 3.2 Modeling Annual Generation

The empirical analysis in Section 4 regresses the price each buyer pays net of the subsidy on the subsidy itself, so it requires the value of the subsidy for each project. Because both the SMART and Solar Carve-Out II programs are production-based subsidies, the total subsidy depends on how much electricity a system will generate and when the payments are made. SMART payments are made monthly through the utility, so the subsidy is the incentive rate times monthly generation. Under the SREC program, payment comes from selling certificates, which are issued every three months, so the subsidy is the market price of certificates times the number of certificates produced each quarter. For simplicity, I calculate the subsidy on an annual basis, which requires annual generation for each project.

I model annual generation using the PVWatts calculator, a tool developed by the National Renewable Energy Laboratory (renamed the National Laboratory of the Rockies in December 2025) for modeling the energy production of grid-connected photovoltaic systems. The calculator combines solar irradiance and meteorological data for a project's location with a set of system parameters to compute monthly generation.

The most important inputs for the calculator are a project's size and location, which

[11] Winsorizing the price per kW at the 1st and 99th percentiles instead of dropping these projects leaves the estimates essentially unchanged: 0.750 (Solar Carve-Out) and 1.016 (SMART) in the naive specification and 0.855 and 0.838 in the preferred specification, compared with 0.756, 1.017, 0.863, and 0.847.

determine the area of solar panels and the available solar resource. The public DOER data contain only ZIP codes, which can cover large areas relative to the resolution of the solar resource data, so I obtained each project's street address from DOER and converted it to coordinates (Appendix A). Figure 2 shows the location of each project in the Solar Carve-Out and SMART samples. The main difference between the two samples is coverage: households in municipal utility territories are ineligible for SMART.[12] In both panels, the density of projects closely resembles the population distribution of Massachusetts.

Besides size and location, other required parameters include the DC-to-AC ratio, module type, array type, azimuth, and tilt.[13]

The calculator output includes monthly and annual generation for each project. I summarize it with the capacity factor: a system's annual generation as a share of what it would produce if it ran at full rated capacity all year. Most projects have modeled capacity factors of 12.5–15%, and a few are much lower.[14]

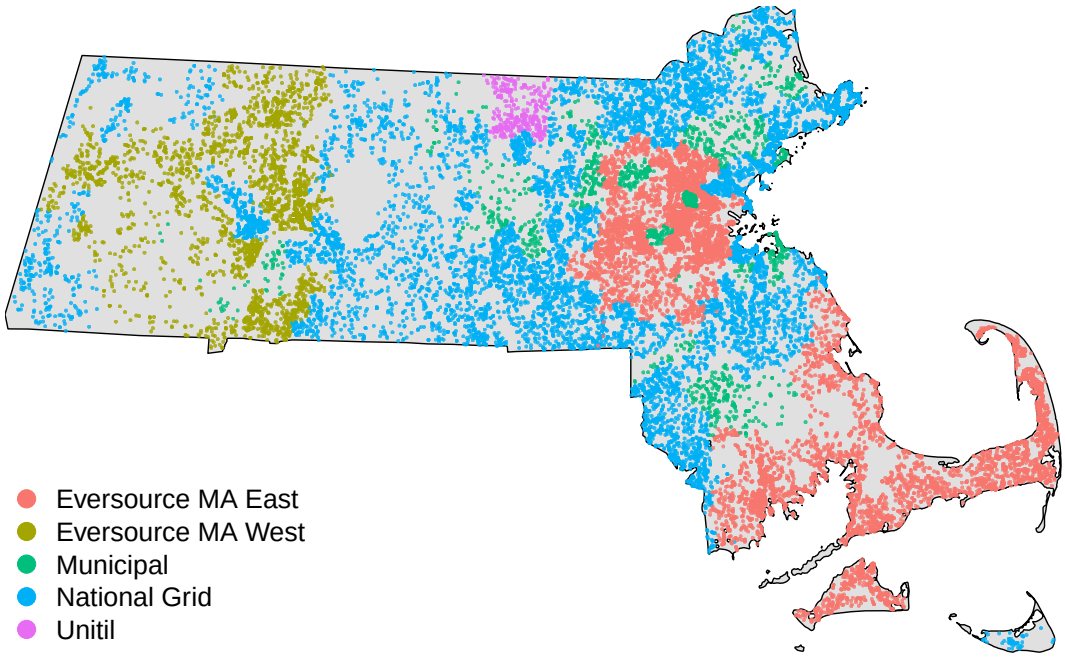
## 3.3 Validating Modeled Generation

The subsidy is proportional to modeled generation, so its accuracy affects both the regressor and the dependent variable. PVWatts output is a model prediction, not a measurement. I fix the system loss parameter at the calculator default of 14.08%, assume a fixed roof-mount array with standard modules, and use observed size, tilt, azimuth, and DC-to-AC ratio where available. The calculator does not model site-specific losses from shading, snow cover, soiling, and unavailability beyond the default values.

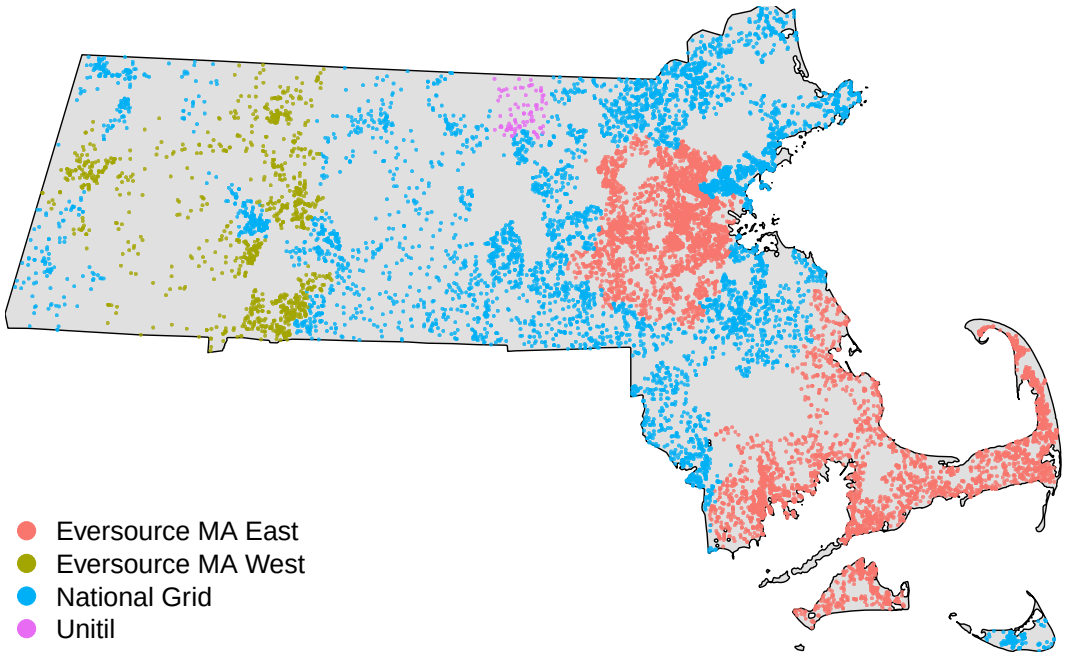
[12] Excluding the 1,567 Solar Carve-Out projects in municipal utility territories changes the Solar Carve-Out estimates little: 0.741 in the naive specification and 0.852 in the preferred specification, compared with 0.756 and 0.863.

[13] I enter the coordinates as the location input. System size is available for every project, but other parameters are missing for some projects. For a missing parameter, I calculate an implied value. For example, I compute the DC-to-AC ratio for SMART projects from their DC and AC ratings. When no implied value is available, I use the calculator's default values, which represent typical rooftop systems. Fewer than 100 projects require default values.

[14] By entering the technical specifications into the calculator manually, I verified that the low values are correct. They come from suboptimal configurations: a north-facing system, for example, generates much less than a south-facing one, all else equal.




(a) Solar Carve-Out II




(b) SMART

Figure 2: Project Locations by Utility Territory. Each dot is one project in the analysis sample. Municipal utility customers are ineligible for SMART.

The average modeled capacity factor is 14.3%, above the 13.1% DOER reports from metered generation for the whole Massachusetts fleet in 2010–2023.[15] The level of modeled generation matters less for the pass-through estimates than this gap might suggest. If PVWatts overstated every project's generation by the same factor $k$, the measured subsidy would be $k$ times the true one. Because the dependent variable subtracts the measured subsidy, the estimated pass-through would then be $1-(1-\rho)/k$, which is closer to one than the true $\rho$. Even if the whole gap of $k \approx 14.3/13.1 \approx 1.09$ applied to my sample, a true pass-through rate of 0.84 would appear as 0.85. Part of the gap comes from older panels in the statewide fleet, so the effect on my estimates is smaller still.

More important is how generation varies across projects. Figure 3 shows that project-level capacity factors carry information that a town or statewide average would miss. The standard deviation across projects is 0.012 around a mean of 0.143, so more than half of projects (53%) differ from the statewide mean by more than 5% and 18% differ by more than 10%. About 91% of the variance is within towns, which means that assigning each project its town average would discard most of the variation in subsidy value across roofs.

Errors in modeled generation that vary across projects pull the pass-through estimate toward one, not toward zero, because the measured subsidy enters both sides of the regression.[16] Errors in the inputs, such as a misrecorded tilt or azimuth or a geocoding error, are of this kind. Losses that PVWatts does not model and that are unrelated to its inputs, such as shading from a neighbor's tree, are not: if true generation equals modeled generation plus an independent error, the estimate is unbiased.

A second measure of generation gives a sense of how large the first kind of error is.

[15] Panel degradation explains only part of this gap. Field evidence suggests that panels lose about 0.5% of output per year (Deline et al., 2024), and applying that rate over the age distribution of the fleet closes roughly a fifth of the difference. The larger part of the gap most plausibly reflects losses that PVWatts does not model. The default snow loss of zero is unrealistic for Massachusetts, and the default shading and soiling losses of 3% and 2% are likely too low given the long pollen season and restrictions on tree removal. The statewide fleet also includes lower-quality installations that predate my sample.

[16] Write the gross price as $P_i = c_i + (1-\rho)S_i$ and the measured subsidy as $\hat{S}_i = S_i + u_i$. The dependent variable is the net price $P_i - \hat{S}_i$, so with classical error the estimated pass-through is $\hat{\rho} = \lambda\rho + (1-\lambda)$, where $\lambda = \mathrm{Var}(S)/[\mathrm{Var}(S) + \mathrm{Var}(u)]$. Equivalently, in a regression of the gross price on the measured subsidy, the coefficient $1-\rho$ is attenuated toward zero.

Each SMART application reports an estimated annual production for the system, which the applicant, usually the installer, computes with its own design tools. The estimate is available for every SMART project, and for 99% of them it lies within 50% of the PVWatts value (median ratio 0.94). Across all projects, the subsidy per kW computed from the two measures has a correlation of 0.97, because both share the SMART rate. Within the cells of the preferred specification, the correlation of the two capacity factors is only 0.57. If the errors in the two measures are independent, the share of the within-cell variation in the modeled capacity factor that is true variation is 0.57, and for the modeled subsidy, which also varies with the SMART rate, it is 0.80. If the errors are positively correlated, for example because both measures use the same recorded tilt and azimuth, these shares are upper bounds. The applicant's estimate is not a cleaner measure, however. It is prepared for the sale, it is computed with tools and assumptions that differ across installers, and its error can be correlated with the price: an installer that overstates output may also charge more. It is also unavailable for Solar Carve-Out. The analysis therefore measures generation with PVWatts in both programs, so that the subsidy is built the same way for every roof, and Section 7.2 uses the applicant's estimate only as a sensitivity check.

## 3.4 Subsidy Calculation

I first calculate the net present value of the subsidy for SMART projects, assuming a discount rate of 8%. Because the SMART compensation rate is locked in for ten years when the application is approved, the owner's compensation in a given year depends only on that year's generation. De Groote and Verboven (2019) find that system owners heavily discount the future cash flows from such subsidy programs, so I also report results for other discount rates. I follow a standard discounted cash flow model:

$$\mathrm{NPV}_i = \sum_{t=1}^{10} \frac{r_i \times g_i (1-a)^t}{(1+d)^t}$$

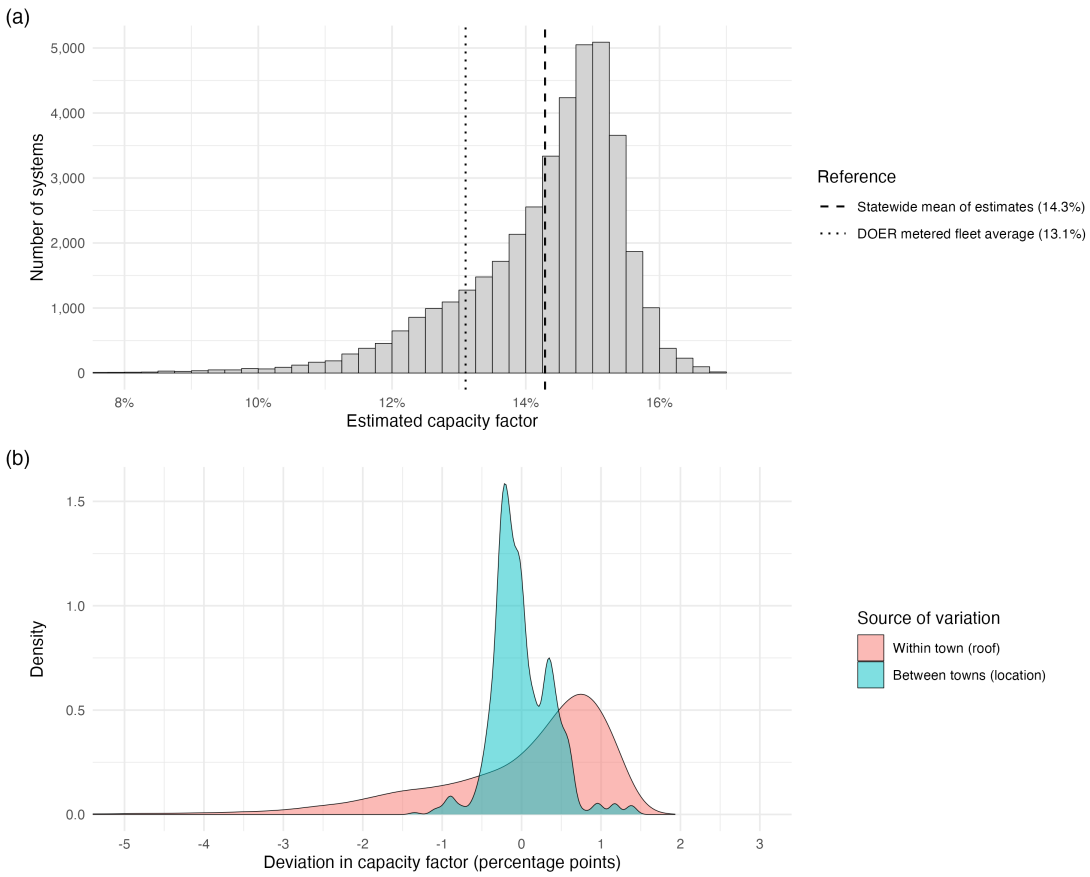



Figure 3: Variation in Modeled Capacity Factor Across Projects

where $r_i$ is the SMART incentive rate, $g_i$ is modeled annual generation, $d$ is the discount rate, and $a$ is the panel degradation rate. I assume annual degradation of 0.5% to account for the performance loss from panel aging. Because $r_i$ and $g_i$ do not change over the ten years, the formula reduces to $\text{NPV}_i = r_i\, g_i\, A(d)$, where $A(d) = \sum_{t=1}^{10}(1-a)^t/(1+d)^t$ is the same for every project. The discount rate therefore rescales every SMART subsidy by the same factor, which is relevant for interpreting the robustness checks in Appendix B.1.

Because SREC compensation rates were market-determined, they are not listed in the Solar Carve-Out data, and the price an owner would receive was not known at the time of purchase. In the initial years, market prices for SRECs were well below the ACP and even consistently below the auction price, so using either the ACP or the auction price alone to proxy for the compensation rate might misstate it. In the main specification, I use the

midpoint of the ACP and the auction price in each year, which is the price a buyer would expect without a view on where the market would settle between the published ACP and auction prices. As an alternative, I use the annual average market price of SRECs, which is the compensation owners actually receive and the price they would expect under rational expectations. For Solar Carve-Out II projects that were still receiving SRECs as of 2025,[17] this alternative uses the average 2024 price for the remaining years. Appendix B.1 compares these and other price assumptions.

For Solar Carve-Out projects, I apply the same approach with certificate prices:

$$\text{NPV}_i = \sum_{t=1}^{10} \frac{g_i \times f_i \times 90\% \times P_t(1-a)^t}{(1+d)^t}$$

where $g_i$ is modeled annual generation in MWh, $f_i$ is the project's SREC factor, $d$ is the discount rate, $a$ is the solar degradation rate, $P_t$ is the SREC price in year $t$, and the 90% term reflects the 10% aggregator fee. I discount the ten years of payments after commercial operation for both programs, and I use the midpoint of the ACP and the auction price as the main price path (Appendix B.1).[18]

## 3.5 Summary Statistics

Table 3 presents summary statistics for installations under the Solar Carve-Out program (Panel A) and the SMART program (Panel B). The largest difference between the programs is in the subsidy per kW. Solar Carve-Out installations received an average subsidy of $1,923 per kW, about 3.4 times the mean SMART subsidy of $566 per kW. As a share of the average gross price per kW, the subsidy is about 45% under Solar Carve-Out and 15% under SMART. This gap reflects the design of the two programs: Solar Carve-Out certificates were priced by a market that tracked a high ACP, whereas SMART rates fall with the value of energy

[17] Projects that began operating after 2015 are still receiving SRECs, but their future prices are not yet known. For example, a 2018 project will keep receiving and selling SRECs until 2028, but SREC prices for 2026 through 2028 are still unknown.

[18] The Solar Carve-Out entitlement is 40 quarters, so ten years of payments covers the whole term.

and as capacity blocks fill.

Table 3: Summary Statistics for SMART and Solar Carve-Out Samples

| Variable | N | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|---|
| *Panel A: Solar Carve-Out Sample* | | | | | |
| Size (kW DC) | 25,491 | 7.96 | 3.20 | 1.10 | 25.00 |
| Total Price ($) | 25,491 | 33,376.11 | 13,130.15 | 5,175.00 | 138,300.00 |
| Price per kW ($/kW) | 25,491 | 4,303.56 | 963.92 | 2,608.75 | 7,847.21 |
| Total Subsidy ($) | 25,491 | 15,169.04 | 6,435.32 | 1,471.30 | 60,419.11 |
| Subsidy per kW ($/kW) | 25,491 | 1,922.77 | 336.09 | 757.12 | 2,631.60 |
| Net Price ($) | 25,491 | 18,207.07 | 8,561.03 | 796.19 | 94,096.40 |
| Net Price per kW ($/kW) | 25,491 | 2,380.79 | 915.74 | 183.45 | 6,300.17 |
| SREC Factor | 25,491 | 0.92 | 0.10 | 0.80 | 1.00 |
| Tilt | 25,491 | 29.39 | 9.13 | 0.00 | 89.00 |
| Azimuth | 25,491 | 181.01 | 41.31 | 5.00 | 356.00 |
| N Inverter | 25,491 | 11.86 | 14.37 | 1 | 280 |
| Annual Generation | 25,491 | 10,062.57 | 4,105.70 | 1,150.13 | 33,998.16 |
| Capacity Factor | 25,491 | 0.14 | 0.01 | 0.05 | 0.17 |
| *Panel B: SMART Sample* | | | | | |
| AC (kW) | 14,317 | 7.49 | 2.64 | 1.20 | 23.50 |
| Size (kW DC) | 14,317 | 9.22 | 3.53 | 1.28 | 24.96 |
| Total Price ($) | 14,317 | 34,305.30 | 14,175.21 | 5,517.00 | 122,655.00 |
| Price per kW ($/kW) | 14,317 | 3,795.74 | 881.11 | 1,998.32 | 7,008.21 |
| Total Subsidy ($) | 14,317 | 5,182.54 | 3,705.16 | 0.00 | 31,917.29 |
| Subsidy per kW ($/kW) | 14,317 | 565.60 | 317.63 | 0.00 | 1,594.76 |
| Net Price ($) | 14,317 | 29,122.77 | 13,216.33 | 3,118.39 | 111,983.60 |
| Net Price per kW ($/kW) | 14,317 | 3,230.14 | 970.62 | 834.28 | 6,976.19 |
| SMART Rate ($/kWh) | 14,317 | 0.07 | 0.04 | 0.00 | 0.20 |
| Tilt | 14,317 | 29.40 | 9.48 | 0 | 84 |
| Azimuth | 14,317 | 179.86 | 56.90 | 0 | 359 |
| N Inverter | 14,317 | 11.57 | 13.52 | 1 | 78 |
| Annual Generation | 14,317 | 11,289.62 | 4,428.34 | 1,547.44 | 34,383.49 |
| Capacity Factor | 14,317 | 0.14 | 0.01 | 0.05 | 0.17 |

Subsidies are present values at an 8% discount rate with 0.5% annual panel degradation. Solar Carve-Out subsidies use the ACP–auction midpoint price path.

The distribution of subsidies also differs. In absolute terms the dispersion is similar. Every Solar Carve-Out system received at least $757 per kW, while some SMART systems received nothing because the value of energy had caught up with the base rate. The maximum subsidy is $2,632 per kW under Solar Carve-Out and $1,595 per kW under SMART, so the two

distributions barely overlap. As a result, a comparison of the two programs is a comparison of very different parts of the subsidy range.

Despite the large difference in subsidy size, both programs supported installations with similar technical characteristics. Average system sizes are comparable: 7.96 kW DC under Solar Carve-Out and 9.22 kW DC under SMART. The mean capacity factor is 0.14 in both samples. The Solar Carve-Out sample has 25,491 systems and the SMART sample 14,317.

# 4 Empirical Strategy

## 4.1 The Naive Model

I estimate the pass-through of solar subsidies to net installation prices with the regression

$$y_i = \alpha + \beta \text{Subsidy}_i + X_i\gamma + \delta_{u(i)} + \delta_{t(i)} + \varepsilon_i$$

where $y_i$ is the net installation price per kW after subsidy (gross price minus subsidy, per kW of DC capacity), $\text{Subsidy}_i$ is the net present value per kW of the subsidy accumulated over ten years, $X_i$ is a set of controls, $\delta_{u(i)}$ is the fixed effect for the utility $u(i)$ that serves project $i$, and $\delta_{t(i)}$ is the fixed effect for the year $t(i)$ in which it began operating. Because $y$ is net of the subsidy, the coefficient $\beta$ equals the negative of the pass-through rate $\rho$: a coefficient of $-1$ means that a dollar of subsidy lowers the net price by a dollar (the gross price does not change), and a coefficient of $-0.76$ means the gross price rises by 24 cents.

The controls $X_i$ include fixed effects for panel manufacturer, inverter manufacturer, installer, and city, all of which affect system prices. Panel and inverter manufacturer fixed effects control for equipment quality, installer fixed effects control for labor costs and installer quality, and city fixed effects control for demographic characteristics and local price levels. Because SMART rates vary across utility territories, utility fixed effects allow me to identify pass-through from changes in the subsidy within a utility territory, where the

compensation structure is the same. Year fixed effects capture the effect of technology and learning by doing on installation costs that is common to a year. I call this model the naive specification, because it uses all of the variation in the subsidy, including its movement within years.

I estimate the model separately for the SMART and Solar Carve-Out samples, using the net present value of the subsidy at a discount rate of 8%. The programs differ in period, in how the subsidy is measured, and in the fixed effects the policy requires, so I let every coefficient differ by program and compare the estimates directly (Appendix B.5 compares this with pooled estimation).[19] All prices and subsidies are in nominal dollars.[20]

## 4.2 Two Sources of Variation

The subsidy per kW varies for two reasons, which differ in how plausibly they are independent of installation costs.

**Timing.** Under Solar Carve-Out, the subsidy per kW varies across installation dates through the SREC price path and, in January 2017, through the cut in the SREC factor. Year fixed effects absorb any level shift common to all projects in a year, but not the movement of the price path within a year or the interaction between the annual price path and the timing of payments (a project installed in 2016 receives a different sequence of vintage prices than one installed in 2018). Under SMART, the base rate falls as capacity blocks fill within a utility territory, and the value of energy differs by utility and over time. Utility fixed effects hold the compensation schedule constant, so the variation comes from movement along the schedule over time.

[19] I do not include utility fixed effects for the Solar Carve-Out sample because SREC prices do not depend on the utility. Adding them leaves the Solar Carve-Out estimate essentially unchanged (Appendix Table 17).

[20] I use nominal dollars for two reasons. First, both are nominal contracts: a SMART rate is fixed in dollars per kWh at approval, SRECs are sold at dollar prices, and a buyer compares the dollar price quoted at purchase with the present value of nominal payments discounted at a nominal rate. Pass-through is a within-period object, the effect of a dollar of subsidy on the price paid on the same date, so there is no economic reason to convert to a common base year. Second, time fixed effects already absorb the economy-wide price level. The choice does affect the naive SMART estimate, because consumer prices rose by about 22% over the SMART sample while rates were stepping down (Section 6.4 and Appendix C.2).

**Generation across roofs.** Subsidy per kW depends on the electricity a system will produce, so two otherwise identical projects in different locations or with different orientations face different subsidies. This variation exists in both programs and does not depend on the policy calendar, so it cannot move together with installation costs over time. An installer may well charge a buyer with a south-facing roof more because that buyer receives a larger subsidy, but that is the incidence the regression measures, not a threat to it.

## 4.3 Why Timing Variation Is Confounded

Write the gross price of project $i$ as $P_i = c_i + (1-\rho)S_i$, where $c_i$ is the price the installer would charge without the subsidy and $S_i$ is the subsidy. The dependent variable is the net price $y_i = P_i - S_i = c_i - \rho S_i$. After the fixed effects and controls are partialled out, the regression coefficient converges to

$$\hat{\rho} \;\to\; \rho - \frac{\mathrm{Cov}(\tilde{c}, \tilde{S})}{\mathrm{Var}(\tilde{S})}, \tag{1}$$

where tildes denote residuals. Any component of cost that moves with the residual subsidy biases the estimate. When the identifying variation comes from timing, $\tilde{c}$ contains the part of the cost trend that the time controls do not absorb. Costs that fall as the subsidy falls ($\mathrm{Cov} > 0$) bias $\hat{\rho}$ down, and costs that rise as the subsidy falls ($\mathrm{Cov} < 0$) bias it up.

The two programs give opposite signs. Under Solar Carve-Out, the SREC price path and installation prices fell together from 2014 to 2018. Under SMART, the rate fell throughout while gross prices rose after 2020 (Figure 5). Equation (1) therefore predicts the sign of the bias for every design that relies on timing: downward under Solar Carve-Out and upward under SMART. Narrowing the window around a rule-based change in the subsidy shrinks the cost trend within the window but does not remove it, and windows narrow enough to remove it contain few projects and are exposed to bunching before the change. Demand shifts point the same way under SMART: blocks fill faster when demand is strong, so strong

demand raises the gross price while moving buyers to a lower block, which also biases the estimate up.

When the identifying variation comes from differences across roofs at the same time, $\tilde{c}$ contains only cost differences across roofs that are correlated with generation. These are the threat to the cross-roof design, and Section 7 examines them.

Comparing the two designs also requires attention to measurement error. Error in modeled generation pulls $\hat{\rho}$ toward one (Section 3.3), and its share of the remaining variation grows as the fixed effects become finer. A cross-roof estimate is therefore pulled further toward one than a naive estimate. Under SMART, this cannot explain why the naive estimate exceeds the cross-roof estimate, because error pulls an estimate toward one, not above it. Under Solar Carve-Out, it works in the same direction as the predicted timing bias, so a rise from the naive to the cross-roof estimate could reflect either. Section 7.2 gauges its size.

## 4.4 The Cross-Roof Design

The preferred specification removes the timing of the subsidy. It replaces year fixed effects with year-month fixed effects (utility-by-year-month for SMART), which absorb the SREC price path and the timing of SMART block changes, adds installer-by-year and city-by-year fixed effects, and controls for the log of system size and five bins of the DC-to-AC ratio. The fixed effects are additive, so pass-through is identified from differences in the subsidy that remain after removing installer-by-year, town-by-year, and month effects, which come mainly from differences in modeled generation. A stricter version replaces installer-by-year with installer-by-month fixed effects, so that only roofs equipped by the same installer in the same month are compared. More than 90% of projects share an installer and month with another project, and the estimates are nearly identical (Section 5.3). Under SMART, part of the remaining variation also comes from differences in incentive rates among projects that began operating in the same month, because the rate is locked at approval, which is usually months before operation. A further specification instruments the subsidy with the project's

modeled capacity factor, so that only the variation that comes from the roof's orientation, tilt, and location identifies pass-through.

The two designs estimate different objects. The naive specification uses changes in the subsidy common to all buyers, whereas the cross-roof design uses differences across buyers, and the pass-through of a market-wide change can differ from that of a change specific to one buyer (Muehlegger and Sweeney, 2022). Section 8.1 shows that, with log-concave demand and marginal cost that does not fall with volume, market-wide pass-through cannot exceed the pass-through of a roof-specific subsidy. Two further issues affect both designs. The discount rate enters both the dependent variable and the regressor, so the coefficient depends on it mechanically, and I report a range of rates (Appendix B.1). And because installers know the subsidy schedule, a timing-based design cannot separate the response of prices to a subsidy cut from pricing that anticipated the cut.

**Inference.** I cluster standard errors at the installer level. Installers set prices, the same installer appears in many projects and years, and an installer's pricing errors are likely to be correlated across its projects. There are 385 installers in the Solar Carve-Out sample and 267 in the SMART sample, which is enough for standard asymptotic inference.[21] To compare the programs, I estimate both in one stacked regression in which the slope, the controls, and every fixed effect are specific to each program, and I cluster by installer across programs. The point estimates equal those from separate regressions, and the standard error of the difference accounts for the 83 installers that operate in both programs, who installed 56% of Solar Carve-Out projects and 69% of SMART projects.

[21] Clustering at the utility level is not appropriate: with only four utility groups in the SMART sample the cluster count is too small, and in the Solar Carve-Out sample there is no utility-based policy variation. Appendix Table 19 shows how the standard errors change under alternative clustering choices. The installer-clustered errors are larger than those clustered by town in both samples, and the conclusions do not change.

# 5 Pass-Through under the Two Programs

## 5.1 Cross-Roof Estimates

Table 4 reports pass-through estimates from specifications that progressively remove the variation in the subsidy that comes from its timing. Column (1) is the naive specification. Column (2) replaces year fixed effects with year-month fixed effects, interacted with utility for SMART. Column (3) adds installer-by-year and city-by-year fixed effects. Column (4) adds the log of system size in kW DC and indicators for five bins of the DC-to-AC ratio and is the preferred specification. Column (5) instruments the subsidy with the project's modeled capacity factor in the specification of column (4).[22]

Table 4: Pass-Through from Cross-Roof Variation in Generation

| | Dependent variable: net price per kW | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| *Panel A: Solar Carve-Out* | | | | | |
| Subsidy per kW | −0.756*** | −0.799*** | −0.826*** | −0.863*** | −0.856*** |
| | (0.044) | (0.046) | (0.044) | (0.044) | (0.050) |
| *Panel B: SMART* | | | | | |
| Subsidy per kW | −1.017*** | −0.887*** | −0.839*** | −0.847*** | −0.894*** |
| | (0.065) | (0.062) | (0.070) | (0.073) | (0.115) |
| Difference in pass-through (SMART − Solar Carve-Out) | 0.260*** | 0.087 | 0.013 | −0.016 | 0.038 |
| | (0.090) | (0.080) | (0.080) | (0.085) | (0.114) |
| Time fixed effects | Year | Year-month | Year-month | Year-month | Year-month |
| Installer × year, city × year | | | ✓ | ✓ | ✓ |
| System size, DC/AC ratio | | | | ✓ | ✓ |
| Subsidy instrumented with capacity factor | | | | | ✓ |

Panels A and B report the coefficient on subsidy per kW. Pass-through is minus this coefficient. The row below the panels reports the difference in pass-through between the programs. All columns include installer, inverter, manufacturer, and city fixed effects (and utility fixed effects for SMART), or their interactions as listed. Discount rate 8%. Observations: 25,491 Solar Carve-Out (23,503 in columns 4 and 5) and 14,317 SMART. Standard errors clustered by installer in parentheses. The difference and its standard error come from one stacked regression with program-specific slopes, controls, and fixed effects, clustered by installer across programs. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

In the preferred specification, column (4), pass-through is 0.86 under Solar Carve-Out and 0.85 under SMART. The difference between the programs is −0.02 with a standard error of 0.085. Both estimates are high but significantly below one. With modeled generation,

[22]The first-stage Wald statistics are 3,661 (Solar Carve-Out) and 1,298 (SMART). Columns (4) and (5) drop 1,988 Solar Carve-Out projects with missing AC capacity.

buyers capture about 85 cents of each dollar of subsidy, whether the subsidy is paid through certificates sold at an uncertain market price or through a fixed rate known at approval. Columns (3) and (5) give the same picture, with differences of 0.01 and 0.04. Instrumenting with the capacity factor, which removes the residual variation in SMART rates, gives 0.89 (0.12) under SMART. Measurement error in modeled generation pulls both levels toward one, so true pass-through may be somewhat lower, and Section 7.2 gauges how much.

The estimates are in the range of prior estimates for solar rebate programs, 78% to close to 100% for host-owned systems (Pless and van Benthem, 2019; Gillingham and Tsvetanov, 2019; Dong, Wiser and Rai, 2018). Those estimates rely on the timing of the subsidy, and Section 6 shows that in Massachusetts such estimates move by more than 0.1 in either direction depending on how costs moved relative to the subsidy.

## 5.2 What the Comparison Rules Out

A finding of no difference is informative only to the extent that it excludes meaningful differences. In the preferred specification, the 95% confidence interval for the difference between the programs is −0.18 to 0.15. It excludes the naive gap of 0.26, so the cross-roof estimates reject the difference that a comparison of the two programs over time would suggest. Column (3), the version with tilt bins (Table 8), and the version with installer-by-month cells give similar intervals. The rejection depends on precision: the instrumental variables estimates in column (5) and the estimate identified from azimuth alone are less precise, and their intervals reach the naive gap, although their point differences are 0.04 and 0.09.[23]

The programs differ in the form of the payment, in its predictability, and in its size (Section 2). Each difference could plausibly change pass-through, and none of them changes it by an amount the data can detect. A buyer receives about the same share of a dollar of

[23] Accounting for installers that operate in both programs barely changes the standard error of the difference in the preferred specification (0.085, compared with 0.084 if the programs were independent) but raises it in the naive specification, from 0.078 to 0.090. The naive gap remains significant ($p = 0.004$).

subsidy whether it arrives through a certificate market or through a fixed rate, and whether the subsidy covers 45% or 15% of the price. Because the two estimates are nearly identical, a single rate summarizes both programs: one slope with program-specific fixed effects gives 0.86. I nonetheless estimate the programs separately. A pooled slope is a variance-weighted average, with about three-quarters of the weight on Solar Carve-Out, and it imposes the equality that the comparison is meant to test. In the naive specification that equality is rejected, and the pooled rate of 0.84 describes neither program. Appendix B.5 reports the pooled estimates.

### 5.3 Robustness

Appendix B reports further checks, all in the preferred specification. Comparing only roofs equipped by the same installer in the same month, with installer-by-month in place of installer-by-year fixed effects, gives 0.860 (0.046) under Solar Carve-Out and 0.859 (0.077) under SMART, a difference of −0.002 (0.087), and installer-by-town-by-year fixed effects give 0.853 and 0.848 (Appendix Table 18). The Solar Carve-Out estimate barely moves with the assumed SREC price path, from 0.844 with the auction price to 0.878 with the ACP. Both estimates fall as the discount rate rises from 3% to 15%, from 0.890 to 0.821 under Solar Carve-Out and from 0.879 to 0.796 under SMART, and both stay significantly below one at every rate. The change is a mechanical rescaling of the subsidy. Adding utility fixed effects to the Solar Carve-Out model leaves the estimate essentially unchanged, and clustering by town or utility instead of by installer does not change the conclusions. SMART pass-through does not vary with ZIP-code income or installer concentration (Appendix B.8).

## 6 Timing-Based Estimates Follow the Cost Trend

Equation (1) predicts that any estimate identified from the timing of the subsidy is biased down under Solar Carve-Out and up under SMART. This section tests the prediction with

every timing-based estimate the data allow: the naive specification, a regression discontinuity at the cut in the SREC factor, a stacked event study around the step-downs in SMART rates, and alternative treatments of the price level. Table 5 collects them, ordered by the length of the period over which each compares projects.

Table 5: Pass-Through by Source of Variation and Length of the Comparison Window

| Design | Projects compared over | Solar Carve-Out | SMART |
|---|---|---|---|
| Cross-roof, preferred (Table 4, col. 4) | Same installer-year, town-year, and month | 0.86 (0.04) | 0.85 (0.07) |
| other cross-roof versions (tilt bins, azimuth only, installer-by-month) | Same installer-year or installer-month | 0.76–0.86 | 0.85–0.86 |
| Year-month fixed effects (Table 4, col. 2) | Same month | 0.80 (0.05) | 0.89 (0.06) |
| Stacked event study, SMART step-downs (Table 7) | 4 months before to 3 months after | | 1.27 (0.15) |
| Regression discontinuity, SREC factor cut (Table 6) | ±90 days | 1.03 (0.10) | |
| | ±180 days | 0.86 (0.07) | |
| | ±365 days | 0.71 (0.08) | |
| excluding Dec. 2016–Jan. 2017 | ±90 / ±180 / ±365 days | 0.76 / 0.61 / 0.54 | |
| Naive, year fixed effects (Table 4, col. 1) | Same year | 0.76 (0.04) | 1.02 (0.07) |
| Predicted sign of the timing bias (equation 1) | | − | + |

Pass-through estimates with installer-clustered standard errors in parentheses. The regression discontinuity rows use the baseline subsidy measure; the standard errors of the estimates excluding December 2016 and January 2017 are 0.21, 0.10, and 0.10. The other cross-roof versions are from Tables 8 and 18. The predicted sign follows from the co-movement of costs and the subsidy in Figure 5: costs fell with the subsidy under Solar Carve-Out and rose as it fell under SMART. Discount rate 8%.

Figure 4 plots the same estimates against the length of the period over which each design compares projects. The shaded band is the range of cross-roof estimates across the versions in Table 5. Under SMART, the naive estimate and the event study lie above the band and at or above one. Under Solar Carve-Out, the estimates scatter around and below the band, and the regression discontinuity falls as its window widens.

## 6.1 The Naive Gap

Column (1) of Table 4 gives pass-through of 0.76 (0.04) under Solar Carve-Out and 1.02 (0.07) under SMART, a gap of 0.26. Complete pass-through is rejected for Solar Carve-Out but not for SMART. These estimates are robust to the choice of controls: Appendix Table 24 shows that adding equipment, installer, and city fixed effects moves them little. Taken at face value, the naive estimates would suggest that a fixed-rate subsidy is passed through more fully than a certificate subsidy.

The SMART base rate is not tied to installation costs: it steps down on a fixed schedule

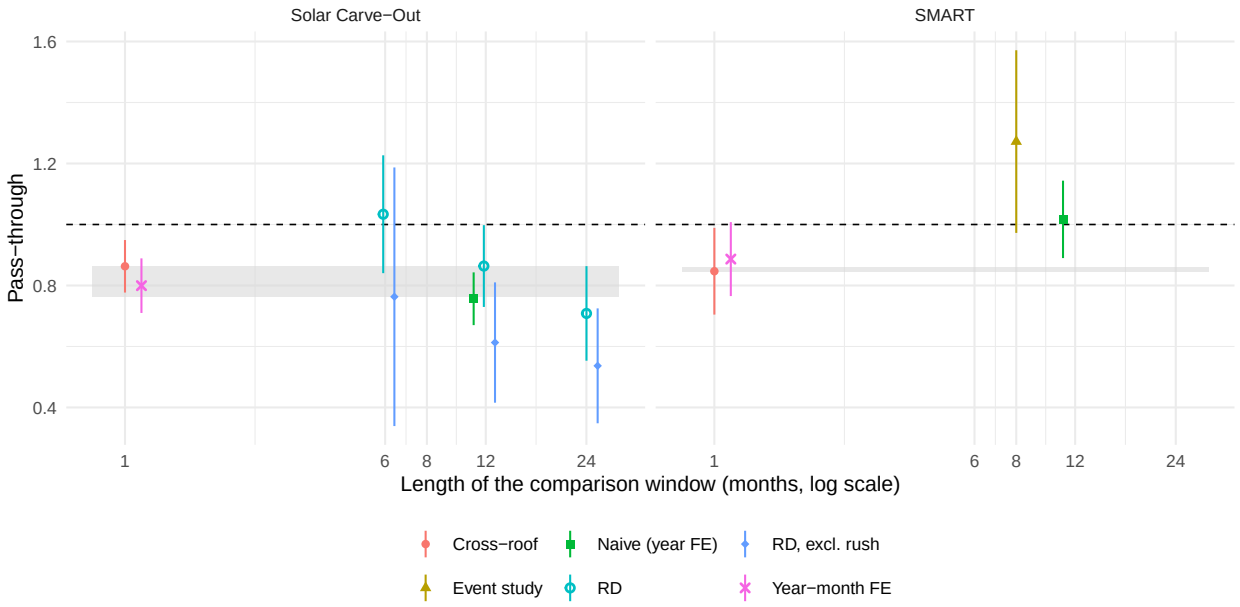



Figure 4: Pass-Through and the Length of the Comparison Window. Each point is an estimate from Table 5, plotted against the length of the period over which the design compares projects (log scale), with a 95% confidence interval. Cross-roof and year-month estimates compare projects within a month, the event study within eight months, the regression discontinuity within ±90, ±180, and ±365 days, and the naive model within a year. The shaded band is the range of cross-roof estimates: column (4) of Table 4, tilt bins, azimuth only, and installer-by-month cells. The dashed line marks complete pass-through.

as cumulative capacity fills the blocks, and the incentive also falls when the value of energy rises. Because blocks fill over time, however, the subsidy changes during the same period in which installation costs change, and the SREC price path co-moves with market conditions. Figure 5 shows the two paths. Under Solar Carve-Out, the subsidy and the gross price fell together from 2014 to 2018. Under SMART, the subsidy fell throughout, from $989 per kW in the first half of 2019 to $329 in the first half of 2023, while the gross price fell until 2020 and then rose from $3,639 to $4,006 per kW by the first half of 2022, as consumer prices rose. The same pattern holds within years: after removing year means, the monthly mean gross price and subsidy are positively correlated under Solar Carve-Out and negatively correlated under SMART. These are the signs of $\mathrm{Cov}(\tilde{c}, \tilde{S})$ in equation (1) that predict a downward bias under Solar Carve-Out and an upward bias under SMART, and they are the directions in which the naive estimates differ from the cross-roof estimates.

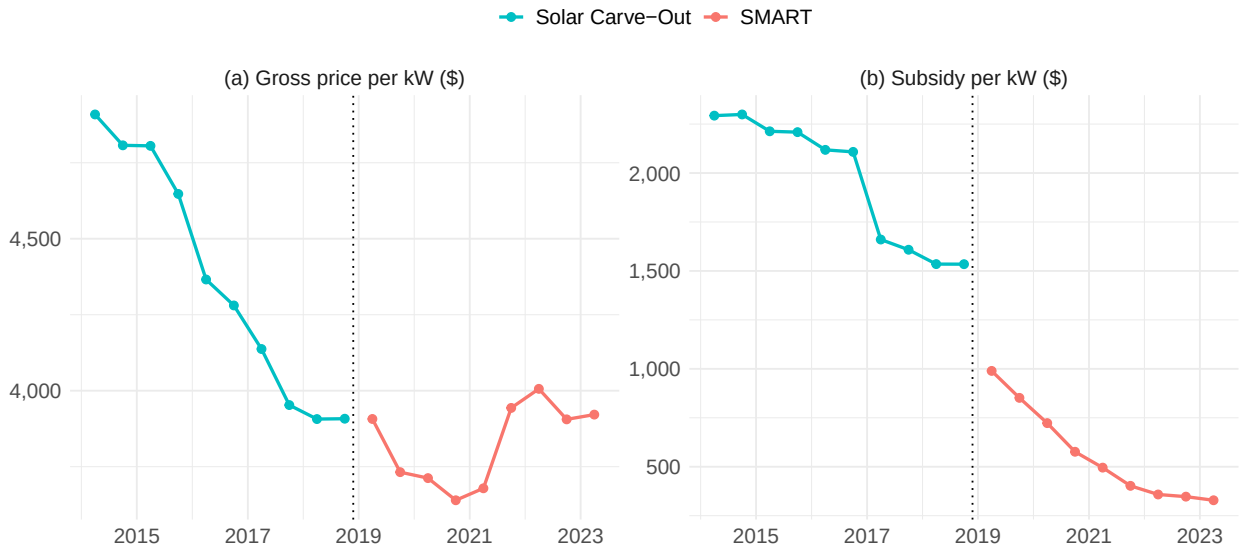



Figure 5: Gross Price and Subsidy per kW over Time. Half-year means of the gross price per kW and the subsidy per kW (present value at an 8% discount rate) by program, without controls. Half-years with fewer than 100 projects are omitted. The dotted line marks the start of SMART.

Year-month fixed effects alone do most of the work: the gap is already down to 0.09 in column (2), before any of the other controls are added. As the timing of the subsidy is

absorbed, the Solar Carve-Out estimate rises from 0.76 to 0.86 and the SMART estimate falls from 1.02 to 0.84–0.89. Most of the naive gap therefore reflects how each subsidy moved over time relative to costs, not how the two designs divide the subsidy. The two movements are not equally informative. The fall in the SMART estimate goes against the pull of measurement error, so the naive SMART estimate is biased up by at least 0.17, and by more if error inflates the cross-roof estimate. The rise in the Solar Carve-Out estimate goes in the direction measurement error would also produce, and the cross-roof estimate identified from azimuth alone, 0.76, equals the naive estimate. The data therefore do not show that the naive Solar Carve-Out estimate is biased, even though costs fell with the subsidy. The naive gap between the programs comes mainly from the SMART estimate.

## 6.2 A Regression Discontinuity at the SREC Factor Cut

On January 8, 2017, the SREC factor for new small systems fell from 1.0 to 0.8 (Section 2). A system that generates the same electricity receives 20% fewer certificates after the cutoff, so the subsidy per kW falls discretely without any change in the installer's costs. A regression discontinuity in time at this cut is the design one would use to avoid trends. I use the commercial operation date as the running variable, with a linear trend on each side of the cutoff and installer, inverter, manufacturer, and city fixed effects, and I instrument the subsidy with the post-cutoff indicator. Figure 6 plots monthly means by month of operation. The subsidy per kW falls from about \$2,110 in December 2016 to about \$1,610 from February 2017, and every project operating after the cutoff has a factor of 0.8, with a single exception.

Two features of the raw data complicate the comparison. First, gross prices fall throughout 2016–2017, by roughly \$230 per kW per year, which is the cost trend that a design in time has to control. Second, the number of projects operating in December 2016 (879) is well above that in each of the eleven preceding months (358–647), which suggests a rush to complete before the deadline, and December projects have unusually low gross prices (about \$4,180 per kW, compared with \$4,240–\$4,380 in the preceding months). I therefore report

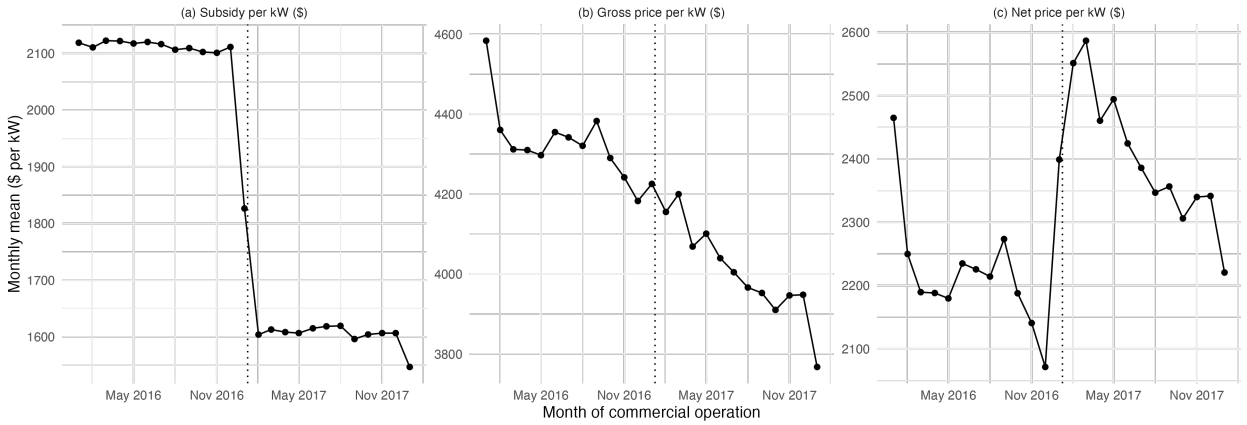



Figure 6: Subsidy, Gross Price, and Net Price by Month of Operation around the SREC Factor Cut

estimates with and without the two months around the cutoff.

Table 6 reports the estimates. The subsidy falls by $465–$500 per kW in the baseline measure and $420–$435 in the factor-only measure, which holds the SREC price path fixed and isolates the change caused by the factor. The estimate depends on the window. With all projects, pass-through falls from 1.03 at a window of ±90 days to 0.86 at ±180 days and 0.71 at ±365 days. Excluding the two months around the cutoff, it falls from 0.76 to 0.61 and 0.54. The wider the window, the more of the falling cost trend enters the comparison and the closer the estimate comes to the naive 0.76, or below it, which is the direction equation (1) predicts. The narrowest windows are imprecise (standard errors of 0.21–0.25 when the rush is excluded) and cannot reject the cross-roof estimates of 0.76–0.86, and the all-project estimate at ±90 days is pushed up by the low December prices, which widen the apparent jump in net prices. No single estimate is decisive, and the discontinuity does not by itself show the sign of the bias under Solar Carve-Out. The evidence is that a design built to avoid trends still gives an answer that moves with the cost trend in its window.

## 6.3 An Event Study around SMART Step-Downs

The same logic applies to SMART: when a capacity block fills, the base rate for the utility falls by a known amount, and the subsidy per kW falls discretely. I identify step-downs from

Table 6: Discontinuity at the January 2017 SREC Factor Cut

| | All projects | | | Excluding Dec. 2016–Jan. 2017 | | |
|---|---|---|---|---|---|---|
| Bandwidth (days) | ±90 | ±180 | ±365 | ±90 | ±180 | ±365 |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Subsidy per kW, baseline measure (first stage) | −466*** | −478*** | −478*** | −495*** | −497*** | −489*** |
| | (12) | (9) | (8) | (34) | (13) | (10) |
| Subsidy per kW, factor-only measure (first stage) | −434*** | −430*** | −424*** | −425*** | −427*** | −422*** |
| | (11) | (9) | (8) | (34) | (13) | (11) |
| Gross price per kW | 16 | −65** | −139*** | −117 | −192*** | −227*** |
| | (46) | (33) | (37) | (104) | (50) | (46) |
| Net price per kW | 482*** | 413*** | 339*** | 378*** | 305*** | 263*** |
| | (48) | (34) | (41) | (114) | (51) | (49) |
| Pass-through $\hat{\rho}$, baseline measure | 1.034*** | 0.864*** | 0.708*** | 0.763*** | 0.613*** | 0.537*** |
| | (0.098) | (0.068) | (0.079) | (0.213) | (0.100) | (0.096) |
| Pass-through $\hat{\rho}$, factor-only measure | 1.110*** | 0.960*** | 0.799*** | 0.888*** | 0.713*** | 0.622*** |
| | (0.105) | (0.076) | (0.089) | (0.246) | (0.117) | (0.110) |
| Observations | 3,206 | 5,995 | 11,795 | 1,722 | 4,527 | 10,328 |

Solar Carve-Out projects. Each column is a set of regressions on projects within the stated window around January 8, 2017, using commercial operation date as the running variable, with a linear trend on each side of the cutoff and installer, inverter, manufacturer, and city fixed effects. Entries for the first four rows are the estimated jump at the cutoff in dollars per kW. Pass-through is estimated by two-stage least squares with the post-cutoff indicator as the instrument for the subsidy. The factor-only measure holds the present-value factor of the SREC price path at its 2016 value, so that the subsidy varies only through the SREC factor and generation. Standard errors clustered by installer in parentheses. Discount rate 8%. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

the monthly modal SMART rate: a step is a month in which the modal rate falls by more than \$0.004 per kWh. Reading the rate this way requires many projects in each month, so I use the two utilities that account for 90% of SMART projects, Eversource East (7,400 projects, 51 months with at least 15 projects) and National Grid (5,747 projects, 46 such months). Unitil has 88 projects and never reaches 15 in a month. Eversource West has 1,082 projects, and its modal rate is noisy in the months with few installations. Adding it gives twelve steps and a pass-through estimate of 1.26 (0.15), almost identical to the estimate below. I keep nine steps that are at least four months from any other step. Each step lowers the rate by \$0.011–\$0.020 per kWh. I stack the nine events and estimate the change in subsidy, gross price, and net price in the four months before and three months after each one, relative to the month before the step.

The data record the date a project began operating, while the SMART rate is fixed when the application is approved, which is usually months earlier. The subsidy therefore does not jump at the step but declines gradually across the window (Figure 7, panel a): it is about \$51 per kW above its month −1 level four months before the step and \$88 below it three months after. If pass-through is complete, the gross price should not respond, and the net price should rise by the amount the subsidy falls. Panel (b) shows that the gross price does not break at the step. Its event-time coefficients lie between −\$40 and +\$26 per kW, with confidence intervals of roughly ±\$40, and panel (c) shows that the net price rises as the subsidy falls.

Table 7 pools the post-step periods. The subsidy is \$85 per kW lower after a step, the gross price is \$23 higher (a standard error of \$13), and the net price is \$108 higher, which implies pass-through of 1.27 (standard error 0.15) when the post indicator instruments the subsidy. The estimate is above one, as the naive SMART estimate is, and in the direction equation (1) predicts for a period in which gross prices drifted up as the subsidy fell. Adding a linear trend in event time yields an estimate of 1.65 with a standard error of 1.43, so once the gradual decline in the subsidy is absorbed by the trend, the design does not identify

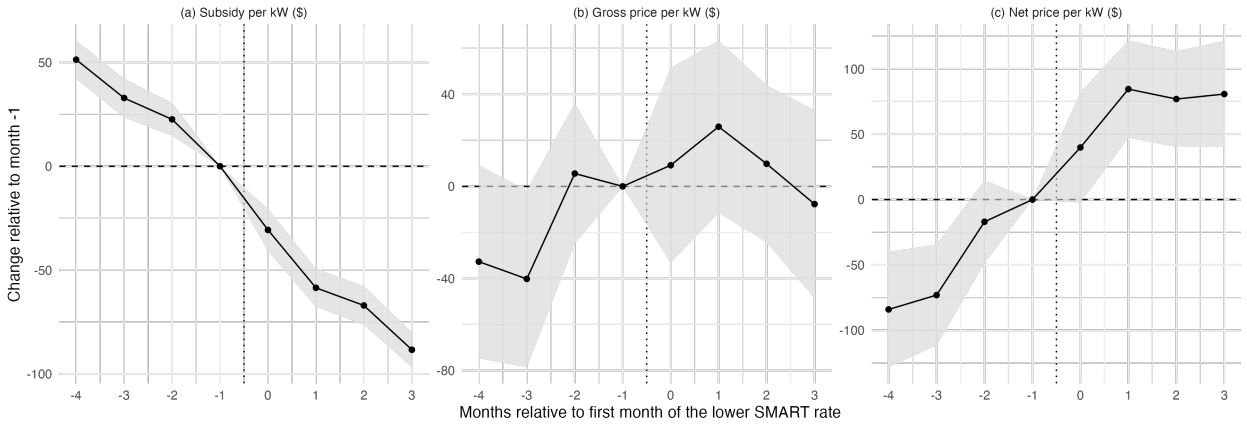



Figure 7: Event Study around SMART Rate Step-Downs

pass-through. Because the rate is locked at approval, the event study compares projects a few months apart rather than across a sharp break. It also rests on nine events in two utilities and cannot rule out that some step-downs coincide with updates to the value of energy.

Table 7: Pooled Event Study around SMART Rate Step-Downs

| | No trend | Linear event-time trend |
|---|---|---|
| | (1) | (2) |
| Subsidy per kW, post vs. pre ($) | −85.2*** | −18.2*** |
| | (2.4) | (5.8) |
| Gross price per kW, post vs. pre ($) | 23.2* | 11.8 |
| | (12.9) | (25.7) |
| Net price per kW, post vs. pre ($) | 108.4*** | 30.0 |
| | (12.8) | (26.1) |
| Pass-through, IV | 1.272*** | 1.645 |
| | (0.153) | (1.435) |
| Observations | 10,851 | 10,851 |

SMART projects. Stacked sample of nine step-downs in the modal SMART rate (two National Grid, seven Eversource East), with projects from four months before to three months after the first month of the lower rate. Post is the indicator for event time $k \geq 0$, and column (2) adds a linear trend in event time. Regressions include event, installer, inverter, manufacturer, and city fixed effects. Pass-through is estimated by two-stage least squares with the post indicator as the instrument for the subsidy. Standard errors clustered by installer in parentheses. Discount rate 8%. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

## 6.4 Other Timing Evidence

Three further results point the same way for SMART. First, the naive SMART estimate is sensitive to the treatment of the price level, because consumer prices rose by about 22% over the SMART sample while rates were stepping down. It is 1.02 in nominal dollars, 0.91 with the CPI as a control, 0.93 with year-by-month fixed effects, and 0.85 in CPI-deflated dollars, while the Solar Carve-Out estimate stays between 0.73 and 0.80 (Appendix C.2). Every alternative moves the SMART estimate toward the cross-roof estimate, because each removes part of the co-movement between the subsidy and costs. Second, by quartile of the subsidy, the naive SMART estimates are 1.55 and 1.32 in the two lowest quartiles, where the subsidy is smallest and the projects are latest (Appendix Table 26). Third, in the naive specification SMART pass-through falls with ZIP-code income, from 1.09 to 0.91 ($p = 0.011$), and this gradient disappears once the timing of the subsidy is absorbed (Appendix B.8).

## 6.5 Market-Wide and Roof-Specific Pass-Through

The timing-based designs and the cross-roof design estimate different objects, so a difference between them need not reflect bias. Under SMART, the timing-based estimates exceed the cross-roof estimate, and with log-concave demand and marginal cost that does not fall with volume, market-wide pass-through cannot exceed the pass-through of a roof-specific subsidy (Section 8.1). Estimates of 1.02 and 1.27 therefore cannot reflect the difference in estimands. Under Solar Carve-Out, the naive estimate lies within the range of cross-roof estimates (0.76–0.86), and the regression discontinuity estimates scatter on both sides of it, so there is no gap for the difference in estimands to explain. A lower market-wide rate would be consistent with a rising supply curve, but under Solar Carve-Out cross-roof pass-through does not fall when an installer or the state is busy (Section 8.3). Appendix C.3 considers three further explanations for the naive gap, the size of the subsidy, its predictability, and competition, and finds that none of them can be separated from the timing of the two programs.

### 6.6 Implications for Existing Estimates

The solar pass-through estimates in the literature use the kind of variation that this section shows to be confounded in Massachusetts. Pless and van Benthem (2019) identify pass-through from "large and sudden jumps in CSI rebate levels" relative to smooth time trends, and their host-owned estimate ranges from 0.70 to 1.01 across the trend specifications in their Table 3. Adding utility-specific trends alone moves it from 0.78 to 1.01, and complete pass-through can then no longer be rejected. Dong, Wiser and Rai (2018) use rebate step-downs within and across California's utilities with ZIP and month fixed effects, Gillingham and Tsvetanov (2019) use Connecticut's statewide step-downs with year fixed effects, and Podolefsky (2013) compares systems six months before and after one change in the federal tax credit with a cubic trend. In each, rebate schedules that fall as deployment grows make the subsidy move with installation costs, and equation (1) applies. I do not re-estimate these studies, and their estimates may well be close to the truth. The point is that their identifying variation, unlike variation across roofs at the same time, does not separate how the subsidy is shared from how costs moved.

## 7 Threats to the Cross-Roof Design

The cross-roof design rests on one assumption: within an installer, town, and period, generation affects the gross price only through the subsidy. This section examines the three ways it could fail.

### 7.1 Roof Traits and Installation Cost

Roofs that generate more could cost more, or less, to install, in ways that system size and the DC-to-AC ratio do not capture, such as pitch, access, or racking. If installation cost changes by $c$ dollars per kW for each percentage point of capacity factor, and the subsidy rises by $s$, equation (1) gives $\hat{\rho} = \rho - c/s$. Within the cells of the preferred specification, the

subsidy rises by $131 per kW per percentage point under Solar Carve-Out and by $40 under SMART, so a cost difference common to both programs would move the SMART estimate about 3.3 times as much as the Solar Carve-Out estimate and would open a gap between them. The equal estimates already argue against a large common $c$.

Within cells, the capacity factor varies mainly with the direction the roof faces. Azimuth bins explain 83% (Solar Carve-Out) and 89% (SMART) of the residual variance of the capacity factor, and tilt bins 9% and 3%. Azimuth determines where the panels point, not how much work the installation takes. Tilt can change cost: flat roofs need tilted racking and steep roofs need more safety work. Appendix Table 21 reports the gross price by tilt bin, holding the subsidy fixed. The only sizable difference is for nearly flat roofs under Solar Carve-Out, which cost $372 (122) more per kW than roofs pitched at 25–30 degrees and generate less. Because those roofs cost more and generate less, they bias the estimate up.

Table 8 removes these differences. Holding tilt fixed with five-degree bins lowers the Solar Carve-Out estimate from 0.86 to 0.82, as the flat-roof premium predicts, and leaves SMART at 0.85. The difference between the programs becomes 0.03 (0.08), with a 95% interval of −0.13 to 0.18. Identifying pass-through only from azimuth, by instrumenting the subsidy with azimuth bins, gives 0.76 (0.04) under Solar Carve-Out and 0.86 (0.12) under SMART, a difference of 0.09 (0.12) that is not significant. Under SMART, every version gives 0.85–0.86. Under Solar Carve-Out, the estimates range from 0.76 to 0.86 depending on which roof traits identify pass-through, so the conclusion that the two programs pass through the same share holds in every version, but the Solar Carve-Out level is less certain than its standard error suggests. The lowest value, from azimuth alone, is what one would expect if part of the remaining variation in modeled generation were error in the other inputs, such as tilt and location. Azimuth is recorded directly on the application, and instrumenting with it removes error in the other inputs that is unrelated to azimuth. Section 7.2 shows that error of this size is plausible.

Table 8: Cross-Roof Pass-Through and Roof Geometry

| | Dependent variable: net price per kW | | |
|---|---|---|---|
| | Cross-roof (preferred) (1) | Tilt bins added (2) | Azimuth only (IV) (3) |
| *Panel A: Solar Carve-Out* | | | |
| Subsidy per kW | −0.863*** | −0.820*** | −0.762*** |
| | (0.044) | (0.035) | (0.039) |
| First-stage $F$ statistic | | | 2942.6 |
| Observations | 23,503 | 23,503 | 23,503 |
| *Panel B: SMART* | | | |
| Subsidy per kW | −0.847*** | −0.845*** | −0.855*** |
| | (0.072) | (0.072) | (0.121) |
| First-stage $F$ statistic | | | 253.6 |
| Observations | 14,317 | 14,317 | 14,317 |
| Difference in pass-through (SMART − SCO) | −0.016 | 0.026 | 0.093 |
| | (0.085) | (0.080) | (0.116) |

Column (1) is column (4) of Table 4. Column (2) adds five-degree tilt bins, specific to each program. Column (3) instruments the subsidy with fifteen-degree azimuth bins in the specification of column (2), so that only azimuth identifies pass-through. All columns come from one stacked regression with program-specific slopes, controls, and fixed effects, clustered by installer across programs. The first-stage $F$ statistics are computed under homoskedasticity. Discount rate 8%. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

## 7.2 Measurement Error in Modeled Generation

The cross-roof estimates rely on the variation most exposed to measurement error in generation, and classical error in the PVWatts inputs pulls the estimate toward one (Section 3.3). The pull is stronger here than in the naive specification: within the same installer, town, and year, roofs differ less in true generation than across the whole sample, so errors make up a larger share of the remaining variation. Instrumenting the subsidy with the modeled capacity factor (column 5 of Table 4) does not remove this, because the instrument comes from the same PVWatts run.

The applicant's production estimate on each SMART application gives a sense of how large this pull is (Table 9). Within the cells of the preferred specification, the two measures of the capacity factor have a correlation of only 0.57, so a substantial part of the remaining variation in modeled generation may be error. Computing the subsidy from the applicant's estimate instead of PVWatts gives SMART pass-through of 0.77 (0.07), compared with 0.85 with PVWatts on the same projects, and instrumenting the PVWatts subsidy with the applicant-estimate subsidy gives 0.74 (0.08). Complete pass-through is rejected in every version.

I do not use these numbers as the main estimates, for three reasons. First, the applicant's estimate is prepared for the sale, with tools and loss assumptions that differ across installers, and its error can be correlated with the price, which would bias both versions. An installer that overstates output to make a system look attractive may also charge more for it, which would push both estimates down. Second, the instrumental variables version requires the errors of the two measures to be independent, which this correlation would violate. Third, Solar Carve-Out records include no production estimate, so the alternative measure cannot be built consistently for both programs. PVWatts applies one model and one set of loss assumptions to every roof in both programs, which is what the comparison between them requires.

The sensitivity check therefore bears on the level of pass-through, not on the compari-

son. Error in modeled generation pulls the estimates toward one, so true pass-through may be lower than 0.85, which strengthens the conclusion that it is incomplete. Because both subsidies are built from the same model, the error has the same source in both programs. Under Solar Carve-Out, the estimate identified from azimuth alone, which is less exposed to error in the other inputs, is 0.76 (Table 8), close to the SMART values with the applicant's estimate. Section 8.2 shows how the bound on market power changes with the amount of error.

Table 9: Modeled Generation and Applicants' Production Estimates under SMART

| *Panel A: Agreement of the two measures (SMART)* | | |
|---|---|---|
| | Capacity factor | Subsidy per kW |
| Correlation, all projects | 0.472 | 0.965 |
| Correlation within cells of column (4) | 0.565 | 0.831 |
| Reliability of PVWatts within cells, $\lambda$ | 0.570 | 0.795 |
| *Panel B: SMART pass-through by measure of generation* | | |
| | $\hat{\rho}$ | SE |
| PVWatts (column 4, validation sample) | 0.846 | (0.076) |
| Applicant's production estimate | 0.773 | (0.069) |
| PVWatts, instrumented with the applicant's estimate | 0.739 | (0.079) |

Panel A compares PVWatts output with the estimated annual production reported on each SMART application, for the 14,196 projects (99%) whose estimate lies within 50% of the PVWatts value. Within-cell values use residuals after the fixed effects and controls of column (4) of Table 4. $\lambda = \text{Cov}(r_{PV}, r_A)/\text{Var}(r_{PV})$, where $r_{PV}$ and $r_A$ are the residuals of the PVWatts and applicant measures, is the share of the within-cell variation in the PVWatts measure that is true variation if the errors of the two measures are independent. The subsidy per kW varies with the SMART rate as well as with generation. In Panel B, the applicant-estimate subsidy uses the same rate and present-value factor with the applicant's generation. The instrumental variables estimate instruments the PVWatts subsidy with it (first-stage Wald statistic 1,371). Both alternative estimates are sensitivity checks: the applicant's estimate is prepared for the sale and its error may be correlated with the price. Standard errors clustered by installer in parentheses. Discount rate 8%.

## 7.3 Rates Locked at Approval

Under SMART, the rate is locked when the application is approved, while the data record the month the system began operating. Projects that began operating in the same month in the same utility can therefore carry different rates, so part of the SMART variation in column (4) comes from rates rather than generation. These rate differences reflect when a

project applied, which could be correlated with its cost. Column (5) of Table 4 uses only the variation in the subsidy that comes from the modeled capacity factor and gives 0.89 (0.12), close to column (4). The SMART estimate therefore does not depend on the residual rate variation.

# 8 Implications for Market Power

High pass-through is often read as evidence of competition, but the two are not the same (Weyl and Fabinger, 2013; Pless and van Benthem, 2019). This section asks what the cross-roof estimates reveal about installers' market power. The answer is that the market power the subsidy reveals is a limited amount of pricing by roof, which bounds markups under SMART at below about 10% of price in most specifications, and that the other sources of market power the data can measure leave little trace.

## 8.1 What Roof-Specific Pass-Through Measures

Weyl and Fabinger (2013) provide a framework for interpreting the size of pass-through. Consider a symmetric market in which installers' conduct is summarized by a parameter $\theta$, which equals 0 under perfect competition and 1 under monopoly and does not change with the subsidy. The pass-through rate of a per-unit subsidy that changes for all buyers at once is

$$\frac{1}{\rho_M} - 1 = r + \frac{\theta}{\varepsilon_{ms}}, \tag{2}$$

where $r \geq 0$ is the slope of marginal cost relative to the slope of inverse demand and $\varepsilon_{ms}$ is the elasticity of the marginal consumer surplus, which measures the curvature of demand. The term $1/\varepsilon_{ms}$ is zero when demand is log-linear, positive when demand is log-concave (for example, linear), and negative when it is log-convex (for example, constant elasticity).[24]

[24] In elasticities, $1/\rho_M = 1 + (\varepsilon_D - \theta)/\varepsilon_S + \theta/\varepsilon_{ms}$, where $\varepsilon_D$ is the absolute value of the demand elasticity and $\varepsilon_S$ the elasticity of supply. Pless and van Benthem (2019) use the same expression, with an additional term for conduct that changes with the subsidy.

With log-concave demand, market power pushes pass-through below one, so a market-wide pass-through rate above one requires marginal cost that falls with volume.

The cross-roof estimates use a different kind of variation: differences in the subsidy across buyers of the same installer in the same town and month. A higher subsidy for one buyer does not change the installer's volume appreciably, so the supply term drops out, and the pass-through of a roof-specific subsidy is

$$\frac{1}{\rho_B} - 1 = \frac{\theta_B}{\varepsilon^B_{ms}},$$

where $\theta_B$ measures the installer's pricing power over individual buyers and $\varepsilon^B_{ms}$ is the curvature of demand among buyers with similar roofs (Appendix E derives this expression from a pricing condition for each roof type). With log-concave demand and $r \geq 0$, $\rho_M \leq \rho_B$. If installers charged one price per kW regardless of the roof, or if the market were perfectly competitive, then $\rho_B = 1$, however large the installer's markup. Pass-through below one therefore requires that the gross price rise with the roof's subsidy: the shortfall $1 - \rho_B$ is the share of a roof's subsidy that the installer captures by pricing the roof. The distinction parallels Muehlegger and Sweeney (2022), who show that fine fixed effects identify the pass-through of a shock specific to one firm, which can differ greatly from that of a shock common to the industry. A roof-specific subsidy differs from a firm-specific cost shock in one respect: every installer that quotes the roof faces the same subsidy, so competition for the job already reflects it.

Such pricing is a form of price discrimination, and the conditions for it hold unusually well in this market. Installers quote each project individually, and they compute the roof's expected output themselves: every SMART application reports the applicant's estimate of the system's annual production (Section 3.3), so the installer knows the subsidy the buyer will receive. An installation cannot be resold. A production-based subsidy also makes such pricing easier than an upfront rebate does, because it varies with a characteristic the

installer observes. An installer with market power would therefore have both the means and the incentive to charge more for roofs that earn more.

Installers do set individual prices, but they price the roof only a little. Table 10 compares the dispersion of prices with the price gap that roof productivity produces. After the controls of column (4) of Table 4, the gross price per kW varies with a standard deviation of $447 under Solar Carve-Out and $488 under SMART among buyers of the same installer in the same month, so prices are not set from a uniform schedule. A roof that is one standard deviation more productive earns $137 more subsidy per kW under Solar Carve-Out and $59 more under SMART, but it is charged only $20 and $6 more, about 4% and 1% of the residual dispersion of prices. The ratio of the two gaps is $1 - \rho$ in column (5) of Table 4. Pass-through is also nearly the same on more and less productive roofs (Appendix Table 22), and Appendix F discusses the welfare effects of pricing by roof.

Table 10: Price Dispersion and Pricing on Roof Productivity

| | Solar Carve-Out | SMART |
|---|---|---|
| Mean gross price per kW ($) | 4,270 | 3,796 |
| Residual SD of gross price per kW ($) | 485 | 521 |
| within installer and year-month ($) | 447 | 488 |
| SD of modeled capacity factor (pp) | 1.04 | 1.45 |
| Subsidy gap for a 1-SD more productive roof ($/kW) | 137 | 59 |
| Gross-price gap for a 1-SD more productive roof ($/kW) | 20 | 6 |
| | (7) | (7) |
| Gross-price gap as % of residual SD | 4.1% | 1.2% |

Residual standard deviations of the gross price per kW after installer-by-year, city-by-year, year-month (utility-by-year-month for SMART), inverter, and manufacturer fixed effects, log system size, and DC-to-AC ratio bins. The within installer and year-month row uses projects that share an installer and month with at least one other project. The gaps are the coefficients on the modeled capacity factor (percentage points) in regressions of the subsidy per kW and of the gross price per kW with the same controls, multiplied by the standard deviation of the capacity factor. Discount rate 8%. Standard errors clustered by installer in parentheses.

## 8.2 A Bound on Market Power under SMART

Turning the shortfall into a bound on conduct requires the curvature of demand. Following Pless and van Benthem (2019), I estimate demand at the ZIP-code and year level, with installations per 1,000 residents quadratic in the mean net price per watt, ZIP characteristics, and county and year fixed effects, and I instrument the price with the mean subsidy per watt and the county wage in building-equipment contracting (Appendix D.1). Under SMART, demand slopes down, with an elasticity at the mean price of −1.57 in the quadratic form and −2.16 in the log-quadratic form, and it is log-concave: the curvature $1/\varepsilon_{ms}$ is 5.38 with the quadratic in levels and 2.14 with the log-quadratic form, which rejects log-linear demand. Both curvature estimates are only marginally significant ($t$-statistics of about 1.9 and 2.1), and the bound below depends on their sign. Under Solar Carve-Out, demand is not identified, because the subsidy is driven by the statewide certificate price and varies little across ZIP codes within a county and year. I therefore draw structural conclusions only for SMART.

The subsidy instrument raises a concern that Section 6 makes concrete. SMART rates step down as capacity blocks fill, and blocks fill faster where demand is strong, so the mean subsidy in a ZIP code and year can be correlated with demand shocks. Two checks bear on this. County-by-year fixed effects absorb demand shocks common to a county and year, including most of the utility-wide shocks that move blocks, and with them the log-quadratic curvature is almost unchanged (2.13, standard error 1.12; Appendix D.2). An instrument built only from roof geometry, the mean modeled capacity factor of a ZIP code's projects in a year, shifts the subsidy without the rate, but it is too weak to be informative (first-stage Wald statistics below 10), so it cannot replace the subsidy instrument. The bound is therefore conditional on the demand estimates.

Because $r$ is absent, $\theta_B = (1/\rho_B - 1)\,\varepsilon_{ms}$ holds with equality if all of the shortfall below one reflects pricing, and it is an upper bound if part of the shortfall reflects cost differences across roofs. The Lerner index equals $\theta/\varepsilon_D$. Table 11 reports the bound on markups for three values

of $\rho_B$ and six demand estimates. With modeled generation ($\rho_B$ = 0.847, Table 4, column 4), $\theta$ is at most 0.034 with the quadratic demand and 0.085 with the log-quadratic demand, so installer markups are at most 2.1% and 3.9% of price, and 4.1–7.0% with the flexible forms of Appendix D.3. If pass-through is instead measured with the applicant's production estimate, at 0.77 or 0.74 (Section 7.2), the shortfall below one is half again as large or more, and the bound rises to 3.5–11.3% and 4.2–13.7% of price. Markups stay below 10% of price in 15 of the 18 combinations. The calculation uses the curvature of aggregate ZIP-level demand in place of the curvature among buyers with similar roofs. Sampling uncertainty widens the upper end: drawing $\rho$ and $1/\varepsilon_{ms}$ from their estimated distributions at $\rho_B$ = 0.847, the 95th percentile of the bound on $\theta$ is 0.15 with the quadratic demand and 0.35 with the log-quadratic demand, which correspond to markups of about 10 and 16 percent of price.

Table 11: Upper Bound on Installer Markups under SMART

| | Roof-specific pass-through $\rho_B$ | | |
|---|---|---|---|
| | 0.847 | 0.773 | 0.739 |
| Demand estimate | Modeled generation (main) | Sensitivity: | applicant's estimate |
| Quadratic, 2SLS | 2.1 | 3.5 | 4.2 |
| Log-quadratic, 2SLS | 3.9 | 6.3 | 7.7 |
| Quadratic, control function | 5.5 | 8.8 | 10.7 |
| Natural spline, 3 df | 4.6 | 7.4 | 8.9 |
| Natural spline, 4 df | 4.1 | 6.6 | 8.0 |
| Natural spline, 5 df | 7.0 | 11.3 | 13.7 |

Entries are the upper bound on the markup in percent of price, $(1/\rho_B - 1)/(1/\varepsilon_{ms})$ divided by the absolute elasticity of demand, at the point estimates of the curvature and elasticity at the mean SMART price (Tables 27 and 29). $\rho_B$ is column (4) of Table 4, with modeled generation, and the two sensitivity values of Table 9, with the applicant's production estimate (OLS and instrumental variables).

Table 12 shows the same bound as a function of the share $\lambda$ of the remaining variation in the measured subsidy that is true variation. If $\lambda$ is known, $\rho_B = 1 - (1 - \hat{\rho}_B)/\lambda$. Markups exceed 5% of price only if $\lambda$ is below 0.82 with the log-quadratic demand or 0.52 with the quadratic demand, and 10% only if it is below 0.48 or 0.33. Taken at face value, the applicant's estimates in Section 7.2 would put $\lambda$ at about 0.6–0.8, where markups are 3–7% of price.

Table 12: Bound on Market Power under Measurement Error in Generation

| | | Quadratic demand | | Log-quadratic demand | |
|---|---|---|---|---|---|
| $\lambda$ | $\rho_B$ | $\theta$ bound | Markup (% of price) | $\theta$ bound | Markup (% of price) |
| 1.0 | 0.847 | 0.034 | 2.1 | 0.085 | 3.9 |
| 0.9 | 0.830 | 0.038 | 2.4 | 0.096 | 4.4 |
| 0.8 | 0.808 | 0.044 | 2.8 | 0.111 | 5.1 |
| 0.7 | 0.781 | 0.052 | 3.3 | 0.131 | 6.1 |
| 0.6 | 0.745 | 0.064 | 4.1 | 0.161 | 7.4 |
| 0.5 | 0.693 | 0.082 | 5.2 | 0.207 | 9.6 |
| $\lambda$ at which the markup reaches 5%: 0.52 (quadratic), 0.82 (log-quadratic) | | | | | |
| $\lambda$ at which the markup reaches 10%: 0.33 (quadratic), 0.48 (log-quadratic) | | | | | |

$\lambda$ is the share of the variation in the measured subsidy, within the cells of the preferred specification, that reflects true differences in generation. $\rho_B = 1-(1-\hat{\rho}_B)/\lambda$ with $\hat{\rho}_B = 0.847$ (Table 4, column 4). The bound is $\theta \leq (1/\rho_B - 1)\,\varepsilon_{ms}$ and the markup is $\theta/\varepsilon_D$, at the point estimates of the SMART demand curvature and elasticity in Table 27. Taken at face value, the applicant's production estimates in Section 7.2 would put $\lambda$ at about 0.6–0.8.

The same framework gives the division of surplus. The ratio of the gain in consumer surplus to the gain in producer surplus is $I = \rho/[1-(1-\theta)\rho]$, so the buyers' share of the surplus that the subsidy creates is $I/(1+I) = \rho/(1+\theta\rho)$, which equals $\rho$ under perfect competition. The formula requires market-wide pass-through, and because $\rho_M \leq \rho_B$, evaluating it at $\rho_B$ gives an upper bound on the buyers' share. At $\rho_B = 0.847$, buyers receive at most 85% of the surplus if the market is perfectly competitive and at most 82% and 79% at the bounds on $\theta$ of 0.034 and 0.085. Market power at the bound therefore shifts at most a few percentage points of the surplus from buyers to installers, beyond what the shortfall in pass-through already shows. The bound assumes that conduct does not change with the subsidy and that marginal cost does not fall with volume. Economies of scale would loosen it.

The comparison with Pless and van Benthem (2019) is direct, because the demand specification is the same. In California, pass-through of 153% to lessees together with convex demand revealed market power in the third-party-owned segment, while host-owned buyers received 78% of the subsidy. In Massachusetts, host-owned buyers receive about 85% of the subsidy under both programs, and log-concave demand turns pass-through below one into an

upper bound on market power rather than evidence of it. The two results concern different segments, periods, and states, and they are not in conflict.

## 8.3 Other Sources of Market Power

Pricing by roof is one way installers could earn a markup. Others would leave different traces, and Table 13 tests their predictions, with the aim of rejecting them. A uniform markup sustained by few competitors would raise gross prices where installers face fewer competitors and, with log-concave demand, lower pass-through where competition is weaker. Rising marginal cost, which lowers market-wide pass-through without market power, implies lower pass-through when installers are busy. Discrimination on willingness to pay rather than on the roof would make pass-through vary with buyer income.[25]

In the preferred specification, cross-roof pass-through does not fall when an installer is busy: it rises by 0.036 (0.022) per standard deviation of the installer's workload under Solar Carve-Out and by 0.039 (0.025) under SMART. Because a single roof does not move an installer along its marginal cost curve, these interactions test whether busy installers price roofs differently, not whether marginal cost rises, and they do not. Under SMART, pass-through is lower in months of high statewide volume, by 0.073 (0.034) per standard deviation, so installers price the roof somewhat more when demand across the state is strong. Under Solar Carve-Out, statewide volume has no effect (0.014, standard error 0.024). SMART pass-through rises with town concentration, by 0.099 (0.047) per standard deviation, which is the opposite of what market power with log-concave demand would predict, and under Solar Carve-Out it does not change with concentration. Across towns, gross prices are lower where more installers are active, by \$66 per kW per log point ($p = 0.05$), or 1.7% of the mean gross price of \$3,796 per kW. This is small, and it may also reflect demand, since installers

[25] I measure an installer's workload as the log of its installations in the month of operation and the month before, relative to the installer's own mean, and statewide workload as the log of monthly installations relative to the mean of that year, and I interact each, standardized, with the subsidy. For town competition, I regress the gross price per kW on the log number of installers active in the town and year and, separately, on the town-year HHI, for towns with at least ten installations. Because city fixed effects are included, these comparisons are within a town over time.

Table 13: Workload, Competition, and Pass-Through

| | Net price per kW | | | Gross price per kW | |
|---|---|---|---|---|---|
| | Installer workload (1) | Statewide workload (2) | Town HHI (3) | Town competitors (4) | Town HHI (5) |
| *Panel A: Solar Carve-Out* | | | | | |
| Subsidy per kW | −0.863*** | −0.857*** | −0.872*** | | |
| | (0.043) | (0.044) | (0.040) | | |
| Subsidy per kW × installer workload | −0.036* | | | | |
| | (0.022) | | | | |
| Subsidy per kW × statewide workload | | −0.014 | | | |
| | | (0.024) | | | |
| Subsidy per kW × town HHI | | | 0.008 | | |
| | | | (0.030) | | |
| Log number of installers in town-year | | | | 42.5 | |
| | | | | (31.3) | |
| Town HHI | | | | | −95.7*** |
| | | | | | (14.4) |
| Observations | 23,503 | 23,503 | 20,086 | 21,783 | 21,783 |
| *Panel B: SMART* | | | | | |
| Subsidy per kW | −0.847*** | −0.881*** | −0.883*** | | |
| | (0.072) | (0.083) | (0.069) | | |
| Subsidy per kW × installer workload | −0.039 | | | | |
| | (0.025) | | | | |
| Subsidy per kW × statewide workload | | 0.073** | | | |
| | | (0.034) | | | |
| Subsidy per kW × town HHI | | | −0.099** | | |
| | | | (0.047) | | |
| Log number of installers in town-year | | | | −66.1* | |
| | | | | (34.0) | |
| Town HHI | | | | | −0.8 |
| | | | | | (16.3) |
| Observations | 14,317 | 14,317 | 11,346 | 11,346 | 11,346 |

The dependent variable is the net price per kW in columns (1)–(3) and the gross price per kW in columns (4) and (5). Columns (1)–(3) use the preferred specification, column (4) of Table 4, so the interactions are identified from differences across roofs within its cells. The main effects of statewide workload and town HHI are absorbed by its fixed effects. Columns (4) and (5) relate prices to the number of installers and the HHI of a town-year, which vary only at the town-year level that city-by-year fixed effects absorb, so they use the naive fixed effects (installer, inverter, manufacturer, city, and year, and utility for SMART) and compare a town with itself over time. Installer workload is the log of the installer's installations in the month of operation and the month before, minus the installer's mean. Statewide workload is the log of all installations in the month, minus the mean for that year. Town competition is measured in town-years with at least ten installations, and the HHI is on a 0–10,000 scale. Workload and HHI are standardized within each program. The coefficient on subsidy per kW is minus pass-through at the mean, and the coefficient on an interaction is minus the change in pass-through per standard deviation. Gross-price regressions also control for subsidy per kW. Discount rate 8%. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

enter towns where demand is growing. In the preferred specification, SMART pass-through is 0.80, 0.88, and 0.86 from the least to the most concentrated tercile of ZIP-level installer concentration ($p = 0.22$) and 0.84, 0.85, and 0.83 across terciles of ZIP-code income ($p = 0.84$; Appendix B.8). These are correlations rather than causal estimates, because workload and the number of competitors respond to demand, and the concentration measures vary within a range of low concentration (Appendix C.3.3).

Pass-through cannot detect one source of market power: a markup common to all buyers in a market, such as rents that free entry dissipates through customer-acquisition costs. Under constant marginal cost and log-linear demand, such a markup is a fixed dollar amount that does not respond to the subsidy, and pass-through is one even though installers gain from the subsidy because they sell more systems at a positive markup. The evidence above shows that the market power the subsidy reveals is limited. It does not show that installers earn no markup, which only cost data or a structural model of installer pricing could settle.

## 8.4 Solar Carve-Out

The two programs pass through the same share of the subsidy across roofs, and 83 installers, accounting for 56% of Solar Carve-Out projects, operate in both. If conduct and the curvature of demand were similar in the two periods, the SMART bound would apply to Solar Carve-Out as well. The data cannot confirm this, because demand under Solar Carve-Out is not identified. The reduced-form checks do not show pricing that responds to capacity: cross-roof pass-through does not fall with installer workload (0.036, standard error 0.022) or statewide workload (0.014, 0.024). The number of competitors in a town has no relation to gross prices, and prices are lower, not higher, in concentrated town-years (–$96 per kW per standard deviation). A likely reason is the Solarize Mass campaigns, in which a town selected a single installer and offered tiered group prices that fell as more households signed up, which made a town concentrated and cheap at the same time (Clean Energy

States Alliance, 2015).[26]

# 9 Conclusion

I estimate the pass-through of two production-based solar subsidies in Massachusetts, using subsidy measures built from each roof's modeled generation, and I identify it from differences in the subsidy across roofs that the same installer equips at the same time.

First, the two designs pass through the subsidy at the same rate. Pass-through is about 85% under both the certificate-based Solar Carve-Out and the fixed-rate SMART program, and the data rule out the gap that a comparison over time suggests. Error in modeled generation may pull these levels toward one, but it has the same source in both programs. The difference between the programs is close to zero, although one subsidy is more than three times the size of the other and only one of them is known at the time of sale.

Second, estimates that rely on the timing of the subsidy follow the cost trends that accompany it. Solar subsidy schedules are designed to fall as deployment grows, so the subsidy moves with installation costs. Under SMART, prices rose as the subsidy fell, and both the naive estimate and an event study at rule-based step-downs lie at or above one, well above the cross-roof estimates. Under Solar Carve-Out, costs fell with the subsidy, the naive estimate lies within the range of cross-roof estimates, and a regression discontinuity at a rule-based cut moves with the length of its window. The gap that a comparison of the programs over time suggests comes from the timing of the subsidy, not from its design.

Third, the result bounds market power. Under a uniform price a roof-specific subsidy would be passed through in full, so the shortfall below one measures pricing by roof. Installers capture about 15 cents of each additional dollar of subsidy a roof earns. With log-concave demand under SMART, this bounds markups at 2–7% of price, and below 10% unless more than about half of the remaining variation in modeled generation is error, conditional on

[26]For example, Lexington and Bedford selected Astrum Solar in a campaign that ran until June 2014 (Town of Lexington, 2014), and Astrum, as Direct Energy Solar, is the second-largest installer in the 2015 data.

demand curvature that is estimated with limited precision.

Three lessons follow for policy and for research on subsidy design. Comparisons of designs across programs or periods confound the design with the cost trends that accompany it, so they should rely on variation that does not move with the policy calendar, and production subsidies provide such variation because their value differs across roofs. The same concern applies wherever subsidies are scheduled to fall with deployment while costs fall with experience, as for electric vehicles and batteries, and the sign of the bias then depends on how costs moved around each change in the subsidy. In a competitive installer market, a production-based subsidy reaches buyers largely, though not fully, whether it is paid through a certificate market or a fixed rate, so the choice between the two can rest on other considerations, such as cost to ratepayers and risk to owners. Finally, most of the variation in subsidy value is across roofs within the same town, so measures that assign town or statewide averages discard it. Errors in measured generation pull pass-through estimates toward one, and a second measure of generation, such as the production estimates that program applications often record, shows how large that pull may be.

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# A Data Construction

*Merging DOER and Tracking the Sun.* Projects are identified differently in the two sources. I match them on operation date, size, and total cost simultaneously. This gives an almost perfect one-to-one match, with only a few projects matched to the same Tracking the Sun record, and I verified manually that these projects have the same technical details.

*Dates.* For the SMART sample, the DOER data list two dates for each project: the Commercial Operation Date and the Incentive Payment Effective Date. I use the Commercial Operation Date, the earliest date in the data. The SMART rate is fixed earlier, when the application is approved, but the data do not record the approval date (Section 7.3).

*Geocoding.* The PVWatts location input accepts street addresses, ZIP codes, or coordinates. The public DOER data contain only ZIP codes, which can cover large areas relative to the resolution of the solar resource data, so I obtained detailed addresses for each project through a separate data request to DOER. The calculator returned unreliable results for street addresses entered directly, so I converted the addresses to coordinates with the U.S. Census geocoder and used the OpenStreetMap geocoding service for addresses that the Census geocoder could not match. All addresses were geocoded by one of the two services, and I verified that all coordinates lie in Massachusetts.

# B Cross-Roof Estimates: Additional Results

## B.1 SREC Prices, Discount Rates, and Tests of Complete Pass-Through

*SREC prices.* I perform a further robustness check specific to Solar Carve-Out using alternative assumptions about system owners' expectations of SREC value. An owner with rational expectations would perfectly predict future SREC prices and sell SRECs once a year at that year's mean price. However, it is hard for an individual to predict the actual price of SRECs, especially in the initial years after the launch of Solar Carve-Out II. The clearinghouse auction price was intended to support SREC prices, but for a period SRECs consistently traded below it because of oversupply (newly installed capacity quickly surpassed the annual target set by the RPS). The common practice of contracting aggregators or brokers to handle SREC trades makes the actual SREC sale price even more difficult to predict. I therefore consider three ways of predicting SREC prices in addition to actual market prices. Optimistic customers use the ACP—the theoretical upper bound—as the predicted SREC price. Pessimistic customers use the clearinghouse auction price. Neutral customers use the average of the ACP and the clearinghouse auction price, which is the main assumption. A standard 10% transaction fee applies to both optimistic and neutral customers. Pessimistic customers are not subject to any other fees, because the transaction fee is already embedded in the clearinghouse auction price.

Table 14 presents pass-through estimates for Solar Carve-Out projects under different SREC price assumptions, in the preferred specification (column (4) of Table 4). Only the Solar Carve-Out sample is shown, and the results are again very close to each other. The highest estimated pass-through rate comes from using the alternative compliance payment as the SREC price, and the lowest from using auction prices. The ordering is as expected, because the ACP is the upper bound of SREC prices and yields the largest subsidy, while the auction price is the lowest assumed path and yields the smallest. The estimates range

from 0.844 to 0.878. The estimate using actual market prices, 0.857, is close to the midpoint estimate of 0.863. Although I cannot determine owners' true beliefs about the income stream SRECs will actually generate, the results show that the Solar Carve-Out estimate is robust to the assumed SREC price path.

Table 14: Pass-Through under Alternative SREC Price Paths

| Dependent Variable: | Net Price per kW | | | |
|---|---|---|---|---|
| Model: | (1) ACP | (2) Auction | (3) Actual price | (4) Midpoint |
| *Variables* | | | | |
| Subsidy per kW | −0.878*** | −0.844*** | −0.857*** | −0.863*** |
| | (0.039) | (0.050) | (0.046) | (0.044) |
| Installer × year | ✓ | ✓ | ✓ | ✓ |
| City × year | ✓ | ✓ | ✓ | ✓ |
| Year-month | ✓ | ✓ | ✓ | ✓ |
| Inverter, manufacturer | ✓ | ✓ | ✓ | ✓ |
| Log size, DC/AC ratio bins | ✓ | ✓ | ✓ | ✓ |
| Observations | 23,503 | 23,503 | 23,503 | 23,503 |

Preferred specification: column (4) of Table 4. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1. Subsidies are discounted at 8% with 0.5% annual panel degradation.

*Discount rates.* Because the subsidy is a discounted sum of future cash flows, the discount rate affects both the dependent variable ($y_i$) and the regressor ($\text{Subsidy}_i$). De Groote and Verboven (2019) find that when subsidies are delivered through electricity production, consumers apply high discount rates to future benefits. They estimate an average rate of 15% in Belgium, well above my main assumption and above the contemporaneous risk-free euro interest rate, which remained below 1% during their study period. I therefore also use rates of 3%, 5%, and 15%.

Table 15 presents pass-through estimates under varying discount rate assumptions, in the preferred specification. In Panel A, the estimated pass-through rate declines from 0.890 at a 3% discount rate to 0.821 at 15%, and in Panel B from 0.879 to 0.796. The discount rate mechanically changes the size of the regressor and the dependent variable. For SMART the

change is purely mechanical. The subsidy is $r_i g_i A(d)$ (Section 3.4), so changing $d$ multiplies every subsidy by the same factor. The coefficient on the subsidy in the gross price, $1 - \rho$, is divided by that factor, and $(\hat{\rho}_d - 1)A(d)$ is the same at every discount rate. With $A(8\%)/A(3\%) \approx 0.79$, the preferred estimate implies $\hat{\rho}_{3\%} - 1 \approx -0.153 \times 0.79 \approx -0.121$, which is the estimate in the table, and the test of $\rho = 1$ gives the same $p$-value (about 0.035) at every rate by construction. The SMART results therefore carry no information about how buyers discount the subsidy. Under Solar Carve-Out the SREC price path differs across cohorts, but the year-month fixed effects absorb these differences, so within a month the discount rate scales every subsidy by nearly the same factor and the result is close to mechanical as well. At every discount rate, both estimates are significantly below one (Table 16).

Table 16 reports the corresponding tests of complete pass-through. For SMART, the standard error times $A(d)$ is also the same at every rate, and so is the $p$-value. The Solar Carve-Out $p$-values are nearly constant for the same reason.

Table 15: Pass-Through at Alternative Discount Rates

| | Panel A: Solar Carve-Out | | | | Panel B: SMART | | | |
|---|---|---|---|---|---|---|---|---|
| | Dependent Variable: Net Price per kW | | | | Dependent Variable: Net Price per kW | | | |
| | (1) d=3% | (2) d=5% | (3) d=8% | (4) d=15% | (5) d=3% | (6) d=5% | (7) d=8% | (8) d=15% |
| *Variables* | | | | | | | | |
| Subsidy per kW | −0.890*** | −0.880*** | −0.863*** | −0.821*** | −0.879*** | −0.867*** | −0.847*** | −0.796*** |
| | (0.035) | (0.039) | (0.044) | (0.057) | (0.057) | (0.063) | (0.073) | (0.097) |
| Installer × year, city × year | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Inverter, manufacturer | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Log size, DC/AC ratio bins | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year-month FE | ✓ | ✓ | ✓ | ✓ | | | | |
| Utility × year-month FE | | | | | ✓ | ✓ | ✓ | ✓ |
| Observations | 23,503 | 23,503 | 23,503 | 23,503 | 14,317 | 14,317 | 14,317 | 14,317 |

Preferred specification: column (4) of Table 4. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1. 0.5% annual panel degradation. SREC subsidies use the ACP–auction midpoint price path.

## B.2 Solar Carve-Out with Utility Fixed Effects

Table 17 adds utility fixed effects to the preferred Solar Carve-Out specification, column (4) of Table 4. SREC prices do not depend on the utility, so the estimate is essentially

Table 16: Test of $\rho = 1$ by Program and Discount Rate

| | d=3% | d=5% | d=8% | d=15% |
|---|---|---|---|---|
| *Panel A: Solar Carve-Out* | | | | |
| Subsidy per kW | −0.890*** | −0.880*** | −0.863*** | −0.821*** |
| | (0.035) | (0.039) | (0.044) | (0.057) |
| $p$-value, $\rho = 1$ | 0.002 | 0.002 | 0.002 | 0.002 |
| *Panel B: SMART* | | | | |
| Subsidy per kW | −0.879*** | −0.867*** | −0.847*** | −0.796*** |
| | (0.057) | (0.063) | (0.073) | (0.097) |
| $p$-value, $\rho = 1$ | 0.035 | 0.035 | 0.035 | 0.035 |

Coefficient on subsidy per kW from the preferred specifications, column (4) of Table 4. The dependent variable is the net price per kW, and pass-through $\rho$ is minus the coefficient. The $p$-values test complete pass-through ($\rho = 1$, a coefficient of −1), using installer-clustered standard errors. Stars test the coefficient against zero. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

unchanged.

Table 17: Solar Carve-Out Pass-Through with and without Utility Fixed Effects

| | Preferred | With Utility FE |
|---|---|---|
| Subsidy per kW | −0.863*** | −0.862*** |
| | (0.044) | (0.044) |
| Installer × year, city × year FE | ✓ | ✓ |
| Year-month, inverter, manufacturer FE | ✓ | ✓ |
| Log system size, DC/AC ratio bins | ✓ | ✓ |
| Utility FE | | ✓ |
| Observations | 23,503 | 23,503 |

Preferred specification: column (4) of Table 4. Standard errors clustered by installer in parentheses. Discount rate 8%. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

## B.3 Comparisons within Finer Cells

The preferred specification includes installer-by-year, city-by-year, and month fixed effects additively, so it does not require two compared roofs to share an installer, a town, and a month at once. Table 18 replaces installer-by-year with installer-by-month fixed effects, which compare only roofs that the same installer equipped in the same month, and alternatively with installer-by-town-by-year fixed effects. In the installer-by-month version,

94% of Solar Carve-Out and 92% of SMART projects share an installer and month with another project. The estimates barely move, and the difference between the programs is −0.002 (0.087).

Table 18: Cross-Roof Pass-Through within Finer Cells

| | Solar Carve-Out | SMART |
|---|---|---|
| Installer × year (column 4) | 0.863 | 0.847 |
| | (0.044) | (0.073) |
| Installer × year-month | 0.860 | 0.859 |
| | (0.046) | (0.077) |
| Installer × city × year | 0.853 | 0.848 |
| | (0.050) | (0.081) |
| Difference (SMART − SCO), installer × year-month | −0.002 (0.087) | |

Coefficients are pass-through (minus the coefficient on subsidy per kW). Each row replaces the installer-by-year fixed effects of column (4) of Table 4 with the stated fixed effects and keeps the other fixed effects and controls. The difference comes from one stacked regression with program-specific slopes, controls, and fixed effects, clustered by installer across programs. Discount rate 8%. Standard errors clustered by installer in parentheses.

## B.4 Alternative Clustering

Table 19: Standard Errors under Alternative Clustering

| Clustered by | Installer | City | Utility |
|---|---|---|---|
| *Panel A: Solar Carve-Out* | | | |
| Subsidy per kW | −0.863*** | −0.863*** | −0.863*** |
| | (0.044) | (0.028) | (0.068) |
| *Panel B: SMART* | | | |
| Subsidy per kW | −0.847*** | −0.847*** | −0.847*** |
| | (0.073) | (0.061) | (0.071) |

Coefficient on subsidy per kW from the preferred specifications, column (4) of Table 4. The dependent variable is the net price per kW, and pass-through is minus the coefficient. Discount rate 8%. Standard errors in parentheses under each clustering. Utility clustering has only four clusters in the SMART sample and is shown for comparison only. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

## B.5 Pooled versus Separate Estimation

Table 20 compares the separate estimates with pooled ones. Columns (1) and (2) come from one stacked regression in which the slope, the controls, and every fixed effect are specific to each program, clustered by installer across programs. Its estimates equal those from separate regressions, and the difference between the programs, reported below the two panels, has a standard error that accounts for installers active in both. Columns (3) to (6) impose one slope. With program-specific fixed effects (columns 3 and 4), the pooled slope is a weighted average of the two program slopes, with weights proportional to the residual variance of the subsidy in each program. Solar Carve-Out receives two-thirds to three-quarters of the weight because its sample is larger. With fixed effects shared across programs (columns 5 and 6), installer, equipment, and town effects are also assumed to be constant from 2014 to 2023.

Table 20: Pooled versus Separate Estimation

| | Dependent variable: net price per kW | | | | | |
|---|---|---|---|---|---|---|
| | Slope by program | | One slope, program FE | | One slope, shared FE | |
| | Naive (1) | Cross-roof (2) | Naive (3) | Cross-roof (4) | Naive (5) | Cross-roof (6) |
| *Panel A: Solar Carve-Out* | | | | | | |
| Subsidy per kW | −0.756*** | −0.863*** | | | | |
| | (0.044) | (0.044) | | | | |
| *Panel B: SMART* | | | | | | |
| Subsidy per kW | −1.017*** | −0.847*** | | | | |
| | (0.065) | (0.072) | | | | |
| Difference in pass-through (SMART − SCO) | 0.260*** | −0.016 | | | | |
| | (0.090) | (0.085) | | | | |
| Standard error if programs independent | (0.078) | (0.084) | | | | |
| Subsidy per kW, both programs | | | −0.843*** | −0.859*** | −0.897*** | −0.858*** |
| | | | (0.031) | (0.037) | (0.038) | (0.037) |
| Weight on Solar Carve-Out | | | 0.666 | 0.736 | | |

Entries are coefficients on subsidy per kW, except the difference row, which reports the difference in pass-through. Pass-through is minus the coefficient. Naive: the specification of column (1) of Table 4. Cross-roof: column (4). Standard errors clustered by installer across programs in parentheses. The weight on Solar Carve-Out is its share of the residual variance of the subsidy after the fixed effects and controls. Discount rate 8%. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

Pless and van Benthem (2019) pool host-owned and third-party systems, which share one program, period, subsidy schedule, and set of fixed effects. The two programs here

differ in all four, so I let every coefficient differ by program. Four considerations support this choice. First, pooling imposes equal pass-through, which the naive specification rejects ($p = 0.004$). The pooled naive rate of 0.84 is then an average that describes neither program, and its weights reflect sample sizes rather than any policy question. Second, the paper's question is whether the two designs differ, which requires a slope for each. Third, the subsidies are measured differently. The SMART subsidy is a fixed contract, whereas the Solar Carve-Out subsidy depends on an assumed SREC price path, and pooling would let that assumption affect the SMART estimate. Fourth, the fixed effects must differ by program anyway. SMART needs utility-by-time effects because its rates differ by utility, the programs overlap only in 2018, and installer, town, and equipment effects plausibly changed between the two periods. Sharing them raises the naive pooled rate from 0.84 to 0.90, because it lets comparisons across programs, in which the subsidy fell by a factor of 3.4 while costs were falling, enter the estimate.

In the preferred specification, the two program estimates are nearly identical, and every version gives about 0.86. Pooling is then harmless, and 0.86 (0.04) is a fair summary of both programs.

## B.6 Roof Tilt and Installation Cost

Table 21 reports the gross price per kW by five-degree tilt bin, relative to roofs pitched at 25–30 degrees, in the preferred specification with the subsidy per kW held fixed. Section 7.1 uses it.

Table 21: Gross Price per kW by Roof Tilt

| | Panel A: Solar Carve-Out | | | Panel B: SMART | | |
|---|---|---|---|---|---|---|
| Tilt (degrees) | $/kW | SE | Projects | $/kW | SE | Projects |
| 0–5 | 372*** | (122) | 333 | 88 | (67) | 208 |
| 5–10 | 86** | (37) | 407 | 25 | (31) | 322 |
| 10–15 | 6 | (19) | 773 | −28 | (24) | 556 |
| 15–20 | −19 | (18) | 2,598 | 13 | (21) | 1,556 |
| 20–25 | −4 | (16) | 4,210 | −6 | (18) | 2,431 |
| 30–35 | 42* | (21) | 4,804 | 6 | (17) | 2,436 |
| 35–40 | −12 | (19) | 3,518 | −21 | (17) | 2,580 |
| 40–45 | 48* | (27) | 2,045 | 17 | (24) | 1,292 |
| 45–50 | −23 | (55) | 131 | 20 | (69) | 113 |
| 50–55 | −165 | (140) | 16 | 137 | (152) | 18 |
| 55–60 | 162 | (186) | 11 | −61 | (193) | 4 |
| 60+ | 199* | (118) | 10 | 86 | (251) | 6 |
| 25–30 (reference) | | | 4,647 | | | 2,795 |

Coefficients on tilt-bin indicators in a regression of the gross price per kW on the subsidy per kW, the tilt bins, and the controls and fixed effects of column (4) of Table 4. The reference bin is 25–30 degrees. Discount rate 8%. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

## B.7 Pass-Through by Roof Productivity

This test estimates cross-roof pass-through by tercile of modeled capacity factor, using the specification in column (4) of Table 4, with terciles defined within each program. Roofs in the low tercile are the weak market and roofs in the high tercile the strong market. In one regression per program, with a separate slope on the subsidy and a separate intercept for each tercile and common fixed effects, pass-through is nearly the same in all three terciles: 0.98, 0.95, and 0.97 under Solar Carve-Out and 0.87, 0.83, and 0.87 under SMART. Equality

is not rejected ($p = 0.66$ and $p = 0.46$), and the difference between the high and the low tercile is −0.01 (0.04) in both programs. Estimating column (4) separately on each tercile's projects gives point estimates that fall from the low to the high tercile, from 0.95 to 0.81 under Solar Carve-Out and from 0.96 to 0.79 under SMART, but the standard errors are 0.07 to 0.20 and the differences are not significant. The estimates within terciles are higher than those in column (4) because the tercile intercepts remove the variation in generation between terciles. Measurement error in generation, which pulls the estimate toward one (Section 3.3), has a larger effect on the variation that remains. The comparison with applicants' production estimates in Section 7.2 suggests that this effect can be large: within the cells of column (4), the two measures of the capacity factor have a correlation of only 0.57, and within terciles the true variation is smaller still. The data therefore do not show that pass-through differs between the weak and the strong market.

Table 22: Pass-Through by Tercile of Modeled Capacity Factor

| | Dependent variable: net price per kW | | | | | |
|---|---|---|---|---|---|---|
| | One regression | | | Separate regressions | | |
| | Low CF (1) | Middle CF (2) | High CF (3) | Low CF (4) | Middle CF (5) | High CF (6) |
| *Panel A: Solar Carve-Out* | | | | | | |
| Subsidy per kW | −0.983*** | −0.952*** | −0.972*** | −0.947*** | −0.806*** | −0.809*** |
| | (0.069) | (0.059) | (0.058) | (0.075) | (0.162) | (0.198) |
| $p$-value, equal across terciles | | 0.658 | | | | |
| *Panel B: SMART* | | | | | | |
| Subsidy per kW | −0.873*** | −0.825*** | −0.867*** | −0.963*** | −0.830*** | −0.787*** |
| | (0.091) | (0.095) | (0.086) | (0.140) | (0.147) | (0.160) |
| $p$-value, equal across terciles | | 0.459 | | | | |

Coefficient on subsidy per kW by tercile of modeled capacity factor (CF), defined within each program, in the specification of column (4) of Table 4. Pass-through is minus the coefficient. Columns (1)–(3) come from one regression that interacts the subsidy with tercile indicators and includes tercile intercepts and common fixed effects, and the $p$-value is from a Wald test that the three slopes are equal. Columns (4)–(6) estimate column (4) of Table 4 separately on each tercile's projects. Discount rate 8%. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

## B.8 Heterogeneity by Income and Installer Concentration

I examine whether SMART pass-through differs across ZIP codes by median household income and by local installer concentration. Income is the ZIP-level median household income from the American Community Survey (2016–2020 five-year estimates). Concentration is the Herfindahl–Hirschman index (HHI) of installers' shares of SMART projects in the ZIP code. I divide ZIP codes into terciles on each measure. For each measure, I estimate a single regression that interacts the subsidy with tercile indicators and keeps the fixed effects common across groups, and then test whether the three coefficients are equal. Table 23 reports the results.

Table 23: Heterogeneity in SMART Pass-Through

| | Dependent variable: net price per kW | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | ZIP median income tercile | | | | Installer HHI tercile | | | |
| Discount rate | 3% | 5% | 8% | 15% | 3% | 5% | 8% | 15% |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Subsidy per kW × tercile 1 | −0.877*** | −0.865*** | −0.844*** | −0.793*** | −0.841*** | −0.824*** | −0.798*** | −0.730*** |
| | (0.058) | (0.064) | (0.073) | (0.098) | (0.067) | (0.074) | (0.085) | (0.114) |
| Subsidy per kW × tercile 2 | −0.886*** | −0.874*** | −0.855*** | −0.806*** | −0.904*** | −0.894*** | −0.878*** | −0.838*** |
| | (0.063) | (0.070) | (0.080) | (0.107) | (0.059) | (0.065) | (0.074) | (0.099) |
| Subsidy per kW × tercile 3 | −0.864*** | −0.849*** | −0.827*** | −0.769*** | −0.888*** | −0.876*** | −0.857*** | −0.810*** |
| | (0.068) | (0.075) | (0.086) | (0.115) | (0.059) | (0.065) | (0.075) | (0.100) |
| $p$-value, equal pass-through | 0.845 | 0.845 | 0.845 | 0.845 | 0.216 | 0.216 | 0.216 | 0.216 |
| Observations | 14,195 | 14,195 | 14,195 | 14,195 | 14,195 | 14,195 | 14,195 | 14,195 |

SMART projects. Each column is one regression, with the subsidy per kW interacted with tercile indicators. Tercile 1 is the lowest-income tercile in columns (1)–(4) and the least concentrated tercile in columns (5)–(8). Entries are coefficients, so pass-through is minus each entry. All regressions use the preferred specification, column (4) of Table 4: utility-by-year-month, installer-by-year, city-by-year, inverter, and manufacturer fixed effects, log system size, and DC-to-AC ratio bins. Standard errors clustered by installer in parentheses. The $p$-value is from a joint Wald test that the three interaction coefficients are equal. It is the same at every discount rate because the discount rate rescales every SMART subsidy by the same factor (Appendix B.1). The sample has 14,195 projects, compared with 14,317 in the full SMART sample. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

Table 23 uses the preferred specification. Pass-through does not vary with income. At the 8% discount rate, the estimates are 0.84 in the lowest tercile, 0.85 in the middle tercile, and 0.83 in the highest tercile, and the hypothesis that they are equal is not rejected ($p$ = 0.84). The pattern is mechanically the same at every discount rate (Appendix B.1). By concentration, the estimates are 0.80, 0.88, and 0.86 from the least to the most concentrated

tercile, so pass-through is lowest where installers are least concentrated, the opposite of what market power would predict, and the equality test does not reject ($p = 0.22$). In the naive specification, pass-through falls with income, from 1.09 to 0.91 ($p = 0.011$), but this gradient disappears once the timing of the subsidy is absorbed.

Two cautions apply. First, the pattern is not robust to how the fixed effects are estimated. If I estimate the model separately within each tercile, allowing every fixed effect to differ by group, the estimates are 0.74, 0.82, and 1.00 from the lowest to the highest income tercile, and 0.85, 0.94, and 0.79 across concentration terciles, with no monotone pattern by concentration (Figures 8 and 9). The split samples are a third of the full sample with installer-by-year and city-by-year effects estimated within each, so these estimates are imprecise, and I do not put weight on either set of gradients. Second, income and concentration are measured at the ZIP level from SMART records only. Income is an ACS estimate for the ZIP, not the buyer's own income, and the HHI is computed from the same projects used in the regression, so ZIPs with few projects have mechanically high concentration. Terciles are equal-sized groups of projects and not of ZIP codes.

Taken together, the results give no evidence that pass-through differs with local income or that it is lower where installers are more concentrated. The second finding bears on the interpretation of the main results: if pass-through under SMART reflected market power, one would expect it to vary with local competition and, since demand is log-concave (Section D.1), to be lower where competition is weaker. The absence of this gradient is consistent with the view that market power is not the main determinant of pass-through in this market, but the test has limited power, since the HHI measure varies within a range of low concentration statewide (Appendix C.3.3).

Figures 8 and 9 report SMART pass-through when the model is estimated separately within income and concentration terciles, so that every fixed effect can differ by group.

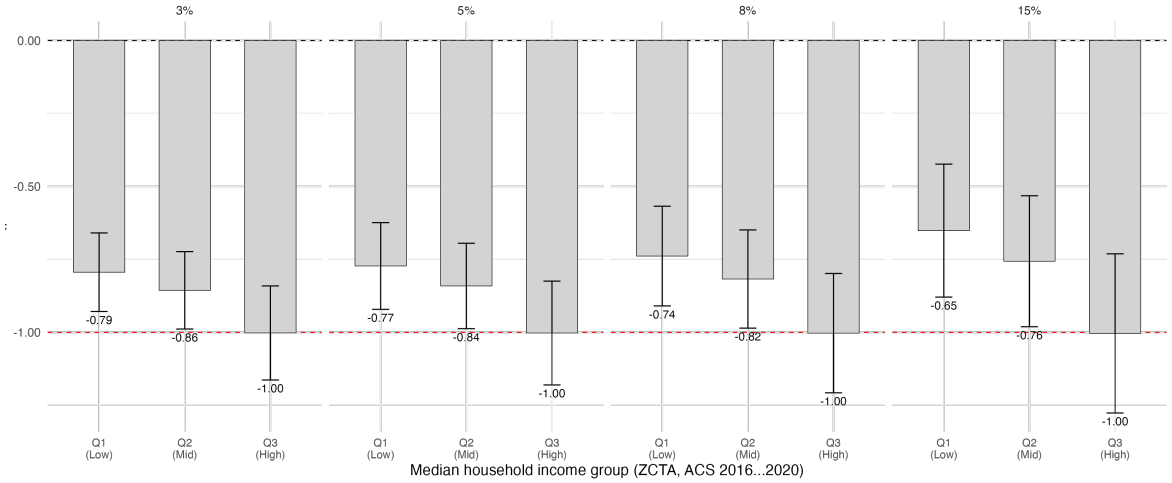



Figure 8: SMART Pass-Through by ZIP Income Tercile, Separate Regressions

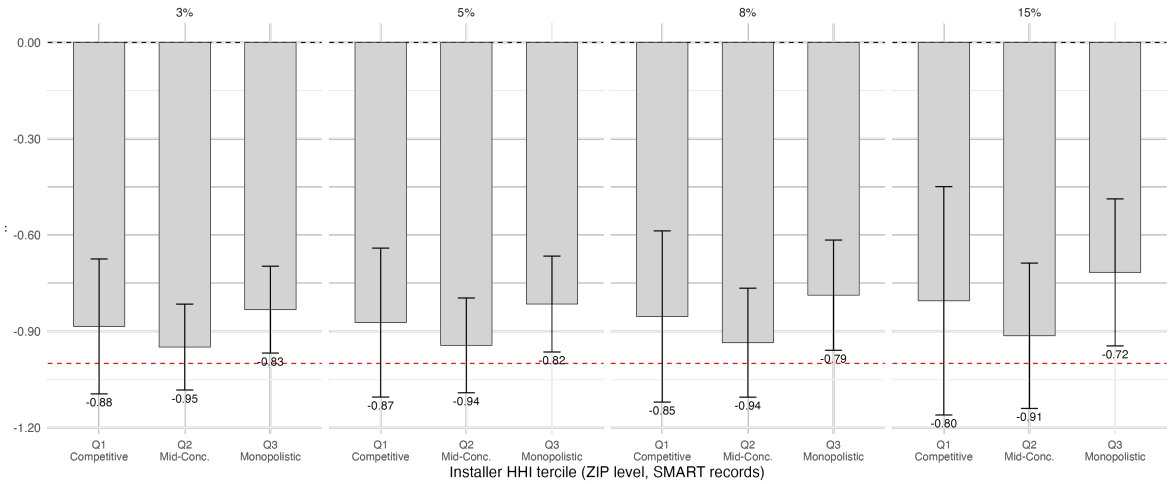



Figure 9: SMART Pass-Through by Installer Concentration Tercile, Separate Regressions

# C Timing-Based Designs: Additional Results

## C.1 Naive Specifications

Table 24 reports the naive model as controls are added. Columns (1) and (5) control only for year fixed effects in Panel A (Solar Carve-Out) and for utility and year fixed effects in Panel B (SMART). Columns (2) and (6) add inverter and panel manufacturer fixed effects. Columns (3) and (7) add installer and city fixed effects. Columns (4) and (8) include all controls and are column (1) of Table 4.

Table 24: Naive Pass-Through Estimates: Solar Carve-Out and SMART

| | Panel A: Solar Carve-Out | | | | Panel B: SMART | | | |
|---|---|---|---|---|---|---|---|---|
| | Dependent Variable: Net Price per kW | | | | Dependent Variable: Net Price per kW | | | |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| *Variables* | | | | | | | | |
| Subsidy per kW | −0.709*** | −0.729*** | −0.751*** | −0.756*** | −0.908*** | −0.888*** | −1.005*** | −1.017*** |
| | (0.109) | (0.073) | (0.042) | (0.044) | (0.138) | (0.089) | (0.061) | (0.065) |
| *Controls* | | | | | | | | |
| Inverter | | ✓ | | ✓ | | ✓ | | ✓ |
| Manufacturer | | ✓ | | ✓ | | ✓ | | ✓ |
| Installer | | | ✓ | ✓ | | | ✓ | ✓ |
| City | | | ✓ | ✓ | | | ✓ | ✓ |
| Utility FE | | | | | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Observations | 25,491 | 25,491 | 25,491 | 25,491 | 14,317 | 14,317 | 14,317 | 14,317 |

Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1. Subsidies are discounted at 8% with 0.5% annual panel degradation. SREC subsidies use the midpoint of the ACP and auction prices.

Across specifications in Panel A, the naive estimates with few or many controls are all close to the full model, so the estimate is not driven by variation in the cost factors that the controls capture. Specifications in Panel B are less similar to one another. Adding manufacturer and inverter controls alone lowers the estimate slightly (0.888), while adding installer and city fixed effects raises it to 1.005, very close to the full model (1.017). The standard errors fall as controls are added, from 0.138 to 0.065 in Panel B, which is consistent with the controls removing cost variation that is unrelated to the subsidy.

## C.2 Price Level

The naive specification is in nominal dollars, for the reasons given in Section 4.1. The SMART estimate is, however, sensitive to how price-level changes are treated: it is 1.02 in nominal dollars, 0.91 when the CPI is added as a control, 0.93 with year-by-month fixed effects, and 0.85 in CPI-deflated dollars (Table 25), while the Solar Carve-Out estimate stays between 0.73 and 0.80. Consumer prices rose by about 22% over the SMART sample while rates were stepping down, so each of these alternatives also removes part of the co-movement between the subsidy and costs discussed in Section 6.1. Because the five specifications use the same projects, I test whether they differ with one stacked regression that gives each specification its own slope, controls, and fixed effects and clusters standard errors by installer across the stacked copies. The SMART estimates differ significantly: every alternative is below the naive estimate, by 0.11 (standard error 0.03) with the CPI control, 0.09 (0.04) with year-by-month fixed effects, 0.17 (0.01) in CPI-deflated dollars, and 0.13 (0.05) in CPI-deflated dollars with year-by-month fixed effects, and equality of all five is rejected ($p < 0.001$). Under Solar Carve-Out the differences are small, from −0.03 to 0.04, but precisely estimated, and equality is also rejected. That the SMART estimate falls whenever part of the rise in the price level is removed is consistent with the naive estimate being pushed up by the timing of the subsidy.

Table 25: Pass-Through under Alternative Treatments of the Price Level

| | Dependent variable: net price per kW | | | | |
|---|---|---|---|---|---|
| | Nominal dollars | | | CPI-deflated dollars | |
| | Year FE (naive) (1) | Year FE, CPI control (2) | Year-by-month FE (3) | Year FE (4) | Year-by-month FE (5) |
| *Panel A: Solar Carve-Out* | | | | | |
| Subsidy per kW | −0.756*** | −0.760*** | −0.799*** | −0.729*** | −0.797*** |
| | (0.044) | (0.046) | (0.046) | (0.042) | (0.045) |
| *Panel B: SMART* | | | | | |
| Subsidy per kW | −1.017*** | −0.906*** | −0.929*** | −0.846*** | −0.883*** |
| | (0.065) | (0.063) | (0.062) | (0.062) | (0.062) |

Full specifications with installer, inverter, manufacturer, and city fixed effects (and utility fixed effects for SMART). Pass-through is minus the coefficient on subsidy per kW. Discount rate 8%. Deflated values use the monthly CPI-U and are expressed in dollars of the first half of 2019 (index 254.412). The CPI ranges from 234 to 253 over the Solar Carve-Out sample and from 248 to 302 over the SMART sample. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

## C.3 Other Explanations for the Naive Gap

The naive estimates differ between the programs: 0.76 for Solar Carve-Out and about one for SMART. Section 6 shows that most of this gap disappears once the timing of the subsidy is absorbed, so the main explanation is how each subsidy moved over time relative to installation costs. This section examines three further candidate explanations. None of the three can be isolated cleanly, because the programs differ in size, certainty, period, and eligible population at the same time, so the evidence here is descriptive. The equal cross-roof estimates suggest that none of them is large.

### C.3.1 Size

The first explanation is that SMART subsidies are small enough to be absorbed without a supply response, whereas Solar Carve-Out subsidies, worth about 45% of the gross price, strain installers' capacity. The mean subsidy is $1,923 per kW under Solar Carve-Out and $566 per kW under SMART, and Section 3 shows that the supports of the two distributions barely overlap. A direct comparison at the same subsidy level is therefore not possible. What I can do is ask whether pass-through falls with the size of the subsidy within each program. Table 26 reports naive pass-through by quartile of subsidy per kW. Under Solar Carve-Out the estimate is flat, between 0.77 and 0.78 in every quartile, so within that program larger subsidies are not passed through less. Under SMART the estimate falls from 1.55 in the lowest quartile to 1.09 in the highest, which is the direction the size explanation predicts. However, the lowest SMART quartile has a median subsidy of $127 per kW and includes projects with no subsidy at all, so the estimate is imprecise (a standard error of 0.23), and the gradient may also reflect rising costs in the later SMART years, when the subsidy was smallest (Figure 5). Estimates above one in the two lowest quartiles are hard to reconcile with log-concave demand (Section 6.5), which favors the second reading. Taken together, the within-program evidence is mixed: it does not support the size explanation under Solar Carve-Out and is consistent with it, but confounded, under SMART. The workload test in

Section 8.3 points the same way for Solar Carve-Out: if large subsidies strained installers' capacity, pass-through would be lower when installers are busy, and it is not.

Table 26: Pass-through by Quartile of Subsidy per kW

| | Q1 | Q2 | Q3 | Q4 |
|---|---|---|---|---|
| *Panel A: Solar Carve-Out* | | | | |
| Median subsidy ($/kW) | 1,528 | 1,698 | 2,151 | 2,305 |
| Subsidy per kW | −0.779*** | −0.776*** | −0.780*** | −0.770*** |
| | (0.073) | (0.068) | (0.062) | (0.059) |
| *Panel B: SMART* | | | | |
| Median subsidy ($/kW) | 127 | 458 | 669 | 947 |
| Subsidy per kW | −1.552*** | −1.317*** | −1.189*** | −1.091*** |
| | (0.228) | (0.131) | (0.091) | (0.076) |

Coefficient on subsidy per kW in each quartile, from a regression of the net price per kW in which subsidy per kW is interacted with quartile indicators of the same variable, with the full set of fixed effects. Pass-through is minus the coefficient. Discount rate 8%. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

### C.3.2 Predictability

The second explanation is that fixed SMART rates reduce uncertainty. Buyers who cannot forecast SREC revenue may value it less than its expected present value, and installers cannot raise prices as far. Three observations bear on this. First, the ACP capped the SREC price and the auction price was meant to support it, so the range of likely prices was narrower than in an unregulated market. Market prices nonetheless fell below the auction price for several years (Figure 1), so the auction price did not act as a floor in practice. Second, the estimate is not sensitive to how one forecasts prices: in the naive specification it ranges from 0.723 (auction price) to 0.782 (ACP) across the three assumptions and is 0.747 with actual market prices, and in the preferred specification it ranges from 0.844 to 0.878 (Table 14). If uncertainty about the price path were the main driver, the estimate should be more sensitive to the assumed path. Third, naive pass-through under Solar Carve-Out falls with the discount rate (0.805 at 3% versus 0.682 at 15%, and 0.890 versus 0.821 in the preferred specification), whereas the SMART estimate cannot respond to

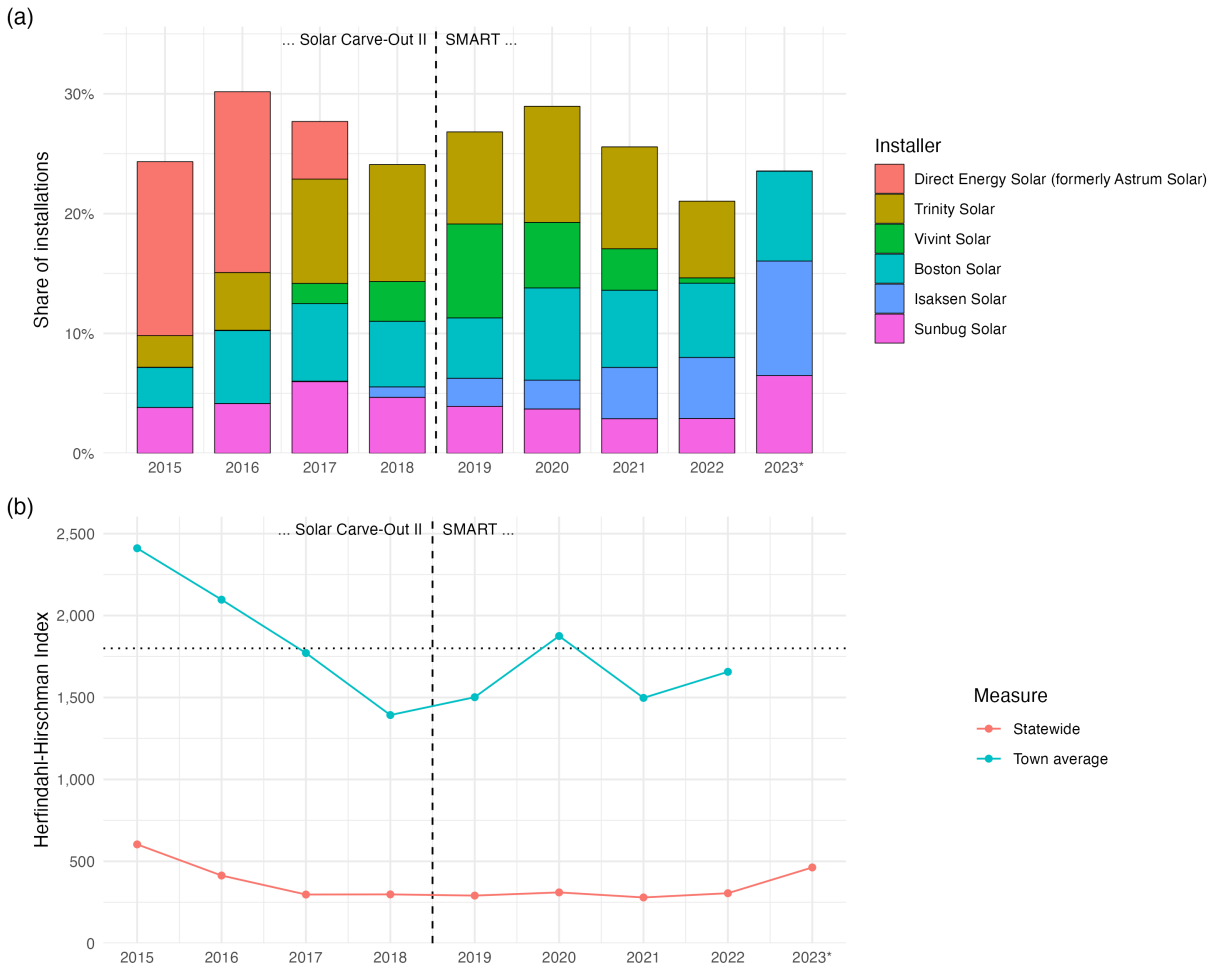



Figure 10: Installer Market Share and Concentration

it by construction (Appendix B.1). The Solar Carve-Out pattern is consistent with buyers applying high discount rates to uncertain, back-loaded certificate revenue, but it is also partly the mechanical effect of a larger regressor.

### C.3.3 Market Competition

The third explanation concerns competition. Figure 10 shows the concentration of installers over the sample. Panel (a) shows the market share of the four largest installers in each program, and panel (b) shows the Herfindahl–Hirschman index (HHI) at the statewide and town levels.

At the state level, the market is unconcentrated throughout. The statewide HHI is 603

in 2015 and falls to about 300 by 2017, where it stays between 280 and 310 through 2022. The top four installers' combined share falls from about 40% in 2015 to 23–27% from 2017 on. The number of active installers per year is 184–216 in 2015–2018 and 128–162 in 2019–2022. The figure starts in 2015 because Solar Carve-Out II opened in April 2014 and, in that partial year, only 28 towns have at least 20 installations, so the 2014 town average (above 6,000) is not comparable with later years. Town-level concentration, which is more relevant for a household choosing among installers who quote locally, is higher: the project-weighted town HHI falls from about 2,400 in 2015 to about 1,400 in 2018 and is 1,500–1,900 in the SMART years with no downward trend within the period.

These patterns do not support the idea that the market became less competitive during SMART, nor do they show that competition intensified in a way that could explain the jump in pass-through. Most of the fall in concentration happened before SMART started, between 2015 and 2017, when Solar Carve-Out pass-through was already about 0.76. Within each program, pass-through rises only slightly with local concentration, and the change is not significant at the 5% level (Section 8.3 and Appendix B.8). Together with low statewide concentration, this suggests that market power is not the main reason for the gap between the two programs.

# D Demand Estimation

## D.1 Main Estimates

The bound in Section 8.2 requires the curvature of demand. I estimate demand following Pless and van Benthem (2019). The unit is a ZIP code and year, and the outcome is installations per 1,000 residents. Demand is quadratic in the mean net price per watt,

$$q_{zt} = \beta_1 p_{zt} + \beta_2 p_{zt}^2 + X_z \gamma + \delta_{c(z)} + \tau_t + e_{zt}, \tag{3}$$

so that $\beta_2$ measures convexity. The vector $X_z$ contains ZIP characteristics from the 2017–2021 American Community Survey (log median household income, the share of adults with a bachelor's degree, median age, log median home value, and the share of owner-occupied homes), $\delta_{c(z)}$ are county fixed effects, and $\tau_t$ are year fixed effects. The net price responds to demand shocks, so I instrument $p$ and $p^2$ with the mean subsidy per watt and its square and with the average weekly wage in the county's building-equipment contracting sector (NAICS 2382, from the Quarterly Census of Employment and Wages) and its square. The subsidy shifts the price buyers pay, and the wage shifts installation costs. Neither should affect adoption except through price. This is the functional form that Pless and van Benthem (2019) use, with the same types of instruments, so a difference in curvature between California and Massachusetts is not an artifact of the specification. Standard functional forms impose the curvature of demand: linear demand is log-concave, exponential demand is log-linear, and constant-elasticity demand is log-convex (Fabinger and Weyl, 2014). I use a quadratic because it is the lowest-order form that estimates curvature rather than imposing it, and I evaluate curvature at the mean price because demand curves need not have constant curvature. Log-concavity is also what one would expect for a discrete adoption decision when willingness to pay has a log-concave density (Bagnoli and Bergstrom, 2005). Appendix D.3 replaces the quadratic with a natural spline. From the estimates, I compute

the price elasticity and the curvature $1/\varepsilon_{ms} = 1 - qq''/q'^2$ at the mean price, with standard errors by the delta method. Because the quadratic in levels imposes a particular shape, I also estimate the same equation with $\log q_{zt}$ as the outcome. In that form the coefficient on $p^2$ is zero exactly when demand is log-linear, so it is a direct test of log-linear demand, and $1/\varepsilon_{ms} = -2\beta_2/(\beta_1 + 2\beta_2 p)^2$. The sample includes ZIP-years with at least one installation and at least 500 residents.

Table 27: Demand Estimates

| | Panel A: Solar Carve-Out | | Panel B: SMART | |
|---|---|---|---|---|
| Dependent variable | Quadratic $q$ (1) | Log-quadratic $\log q$ (2) | Quadratic $q$ (3) | Log-quadratic $\log q$ (4) |
| Net price | 0.777 | 2.110 | 2.516*** | 2.343*** |
| | (4.257) | (2.690) | (0.605) | (0.563) |
| Net price squared | −0.124 | −0.362 | −0.439*** | −0.446*** |
| | (0.716) | (0.459) | (0.082) | (0.075) |
| Price elasticity at the mean | 0.27 | 0.79 | −1.57*** | −2.16*** |
| | (1.21) | (1.09) | (0.42) | (0.42) |
| Curvature $1/\varepsilon_{ms}$ at the mean | 14.86 | 7.05 | 5.38* | 2.14** |
| | (56.01) | (11.35) | (2.77) | (1.02) |
| First-stage Wald statistic, $p$ / $p^2$ | 20.3 / 16.3 | 20.3 / 16.3 | 61.2 / 56.3 | 61.2 / 56.3 |
| Mean net price ($/W) | 2.47 | 2.47 | 3.35 | 3.35 |
| Mean installations per 1,000 residents | 1.54 | 1.54 | 0.91 | 0.91 |
| ZIP-years | 2,171 | 2,171 | 1,715 | 1,715 |

Two-stage least squares estimates. Columns (1) and (3): equation (3), with $q$ the installations per 1,000 residents. Columns (2) and (4): the same equation with $\log q$ as the outcome, so that the coefficient on the squared price is zero under log-linear demand. Instruments: mean subsidy per watt and its square, and the county building-equipment contractor wage and its square. Controls: ACS 2017–2021 ZIP characteristics, county and year fixed effects. $1/\varepsilon_{ms} = 0$ for log-linear demand, $> 0$ for log-concave demand, and $< 0$ for log-convex demand. Standard errors clustered by ZIP code in parentheses. Elasticity and curvature standard errors by the delta method. Appendix D.2 reports alternative specifications. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

Table 27 reports the estimates. Under SMART, demand slopes down with an elasticity of −1.57 at the mean price of $3.35 per watt. This is nearly twice the −0.85 that Pless and van Benthem (2019) report for host-owned systems in California at a mean price of $3.70 per watt.[27] The squared term is negative and precisely estimated, so demand is concave, and the

[27]The difference may reflect a later period in which buyers had more alternatives to buying a system outright, such as leases and power purchase agreements, which are outside my sample. It may also reflect

curvature measure is 5.38. The log-quadratic form gives the same answer more sharply: the coefficient on the squared price is −0.446, which rejects log-linear demand, and the curvature is 2.14. Demand under SMART is therefore log-concave, not log-linear and not log-convex, whereas Pless and van Benthem (2019) find convex demand in California with the same functional form.[28] The result holds across specifications. The elasticity ranges from −1.14 to −1.92 and the curvature from 3.4 to 9.8 when I add utility fixed effects, replace county and year fixed effects with county-by-year fixed effects, weight by population, drop ZIP-years with fewer than three installations, or cluster by county (Appendix D.2). Under Solar Carve-Out, demand is not identified. The subsidy is driven by the statewide certificate price and varies little across ZIP codes within a county and year, and the estimates change sign across specifications. I therefore draw structural conclusions only for SMART.

Two further limitations qualify the SMART result. First, the curvature is measured with limited precision: the quadratic curvature of 5.38 has a $t$-statistic of about 1.9 and the log-quadratic curvature of 2.14 about 2.1, and the bound on market power below depends on its sign. Second, the sample excludes ZIP-years with no installations, which truncates demand at low quantities, and truncation affects the estimated curvature directly.

who is buying: early adopters are more likely to install at almost any price, so later buyers may simply be more price-responsive. Differences in samples and instruments may also contribute.

[28] Both forms bend back at low prices: the level quadratic slopes upward below \$2.87 per watt (a quarter of ZIP-years) and the log-quadratic below \$2.62 (13%), so the estimates describe demand around the mean rather than over the whole range.

## D.2 Alternative Specifications

Table 28 reports the quadratic demand estimates of Section D.1 under alternative specifications. With county-by-year fixed effects and only the subsidy instruments, the log-quadratic SMART estimates are almost unchanged: the coefficient on the squared price is −0.425 (0.103) and the curvature is 2.13 (1.12). Under SMART, the price coefficients and the curvature keep their sign in every specification. Under Solar Carve-Out, the estimates are imprecise and change sign, so demand is not identified. I also replaced the subsidy instruments with the mean modeled capacity factor of each ZIP code's projects and its square, which shift the subsidy only through roof geometry. With the wage instruments, the first-stage Wald statistics are 7.6 and 8.6, and the curvature is positive but has standard errors of 13 (log-quadratic) and more than 1,000 (quadratic). Without them, the first stage has no power. These instruments are too weak to test the subsidy instrument.

Table 28: Demand Estimates under Alternative Specifications

| | Baseline | Utility FE added | County × year FE, subsidy IV | Population-weighted | ZIP-years with 3+ installations | Clustered by county |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A: Solar Carve-Out* | | | | | | |
| Price elasticity at the mean | 0.27 | 0.30 | −2.51 | 2.24 | −4.93** | 0.27 |
| | (1.21) | (1.29) | (1.61) | (1.40) | (2.46) | (1.12) |
| Net price squared ($\beta_2$) | −0.124 | −0.216 | −0.202 | −1.001 | 5.026** | −0.124 |
| | (0.716) | (0.765) | (0.909) | (0.857) | (2.132) | (0.768) |
| Curvature $1/\varepsilon_{ms}$ at the mean | 14.86 | 19.58 | 1.25 | 2.59*** | −0.42 | 14.86 |
| | (56.01) | (97.34) | (1.43) | (0.78) | (0.91) | (58.00) |
| ZIP-years | 2,171 | 2,171 | 2,171 | 2,171 | 1,809 | 2,171 |
| *Panel B: SMART* | | | | | | |
| Price elasticity at the mean | −1.57*** | −1.92** | −1.62*** | −1.21*** | −1.14** | −1.57*** |
| | (0.42) | (0.77) | (0.43) | (0.37) | (0.47) | (0.51) |
| Net price squared ($\beta_2$) | −0.439*** | −0.417*** | −0.251*** | −0.216*** | −0.570*** | −0.439*** |
| | (0.082) | (0.092) | (0.082) | (0.062) | (0.116) | (0.122) |
| Curvature $1/\varepsilon_{ms}$ at the mean | 5.38* | 3.80 | 3.35* | 4.64 | 9.78 | 5.38 |
| | (2.77) | (2.65) | (1.73) | (2.97) | (8.39) | (3.55) |
| ZIP-years | 1,715 | 1,715 | 1,715 | 1,715 | 1,237 | 1,715 |

Two-stage least squares estimates of equation (3). Elasticity and curvature at the mean net price and installation rate of each sample. Baseline: county and year fixed effects, with the subsidy per watt, the county building-equipment contractor wage, and their squares as instruments. Column (3) uses only the subsidy instruments, because the wage varies at the county-year level. Standard errors clustered by ZIP code (by county in column 6) in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

## D.3 Flexible Functional Form

The quadratic in Section D.1 estimates curvature from the data but restricts how curvature varies with price. To check whether this restriction matters, I replace the quadratic with a natural cubic spline in the net price, with three to five degrees of freedom and interior knots at quantiles of the price distribution. A spline enters the endogenous price through several nonlinear terms, and two-stage least squares with polynomials of the instruments identifies them only weakly: with nine instruments, the curvature at the mean ranges from −87 to 1.9 across three to five degrees of freedom, with standard errors of up to 389. I therefore use a control function (Newey, Powell and Vella, 1999). The first stage regresses the price on the instruments, controls, and fixed effects of equation (3), and the demand equation adds the first-stage residual to the spline. Standard errors come from 499 bootstrap draws that resample ZIP codes and re-estimate both stages. Under Solar Carve-Out, the curvature estimates have standard errors in the thousands under every form, consistent with demand not being identified (Section D.1), so Table 29 reports SMART only.

Three findings stand out. First, curvature is positive in every specification, between 0.73 and 1.51 at the mean price, and 97 to 100 percent of bootstrap draws imply log-concave demand. The conclusion that demand under SMART is log-concave therefore does not depend on the quadratic. Second, with the estimator held fixed, the quadratic and the splines give similar curvature (1.31 for the quadratic and 0.73 to 1.51 for the splines), so the quadratic is flexible enough near the mean. Third, the baseline curvature of 5.38 is larger than under any flexible form. Figure 11 shows why. The baseline quadratic turns upward at \$2.87 per watt, not far below the mean of \$3.35, and the curvature of a quadratic rises steeply as its slope approaches zero. From about \$3.80 per watt upward, all three curves in the figure lie between about one and two.

Smaller curvature loosens the bound. At the cross-roof pass-through estimate with modeled generation, the implied markup is at most 4.1 to 7.0 percent of price under the control-function estimates, compared with 2.1 percent for the baseline quadratic and 3.9 percent

for the log-quadratic form. With the sensitivity values of pass-through from the applicant's estimate, 0.77 and 0.74, the control-function bounds are 6.6–11.3 and 8.0–13.7 percent (Table 11). The baseline quadratic gives the tightest bound.

Table 29: SMART Demand Curvature under Flexible Functional Forms

| | Two-stage least squares | | Control function | | | |
|---|---|---|---|---|---|---|
| | Quadratic (baseline) | Log-quadratic | Quadratic | Natural spline | | |
| | | | | 3 df | 4 df | 5 df |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Elasticity at the mean | −1.57*** | −2.16*** | −2.54*** | −3.01*** | −2.94*** | −3.52*** |
| | (0.42) | (0.42) | (0.36) | (0.41) | (0.39) | (0.76) |
| Curvature $1/\varepsilon_{ms}$ at the mean | 5.38* | 2.14** | 1.31*** | 1.32*** | 1.51*** | 0.73 |
| | (2.77) | (1.02) | (0.17) | (0.14) | (0.56) | (0.98) |
| Share of bootstrap draws log-concave | | | 1.00 | 1.00 | 1.00 | 0.97 |
| Markup bound (% of price) | 2.1 | 3.9 | 5.5 | 4.6 | 4.1 | 7.0 |

SMART ZIP-years (1,715). Elasticity and curvature at the mean net price and installation rate. Columns (1) and (2): two-stage least squares from Table 27, with standard errors clustered by ZIP code. Columns (3)–(6): control function, in which the demand equation includes the first-stage residual of the price. Natural splines have interior knots at quantiles of the net price. Standard errors in columns (3)–(6) and the share of log-concave draws come from 499 bootstrap draws that resample ZIP codes and re-estimate both stages. The markup bound is $(1/\rho_B - 1)/(1/\varepsilon_{ms})$ divided by the absolute elasticity, at $\rho_B = 0.847$ (Table 4, column 4). All specifications include the ZIP controls and county and year fixed effects of equation (3). Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

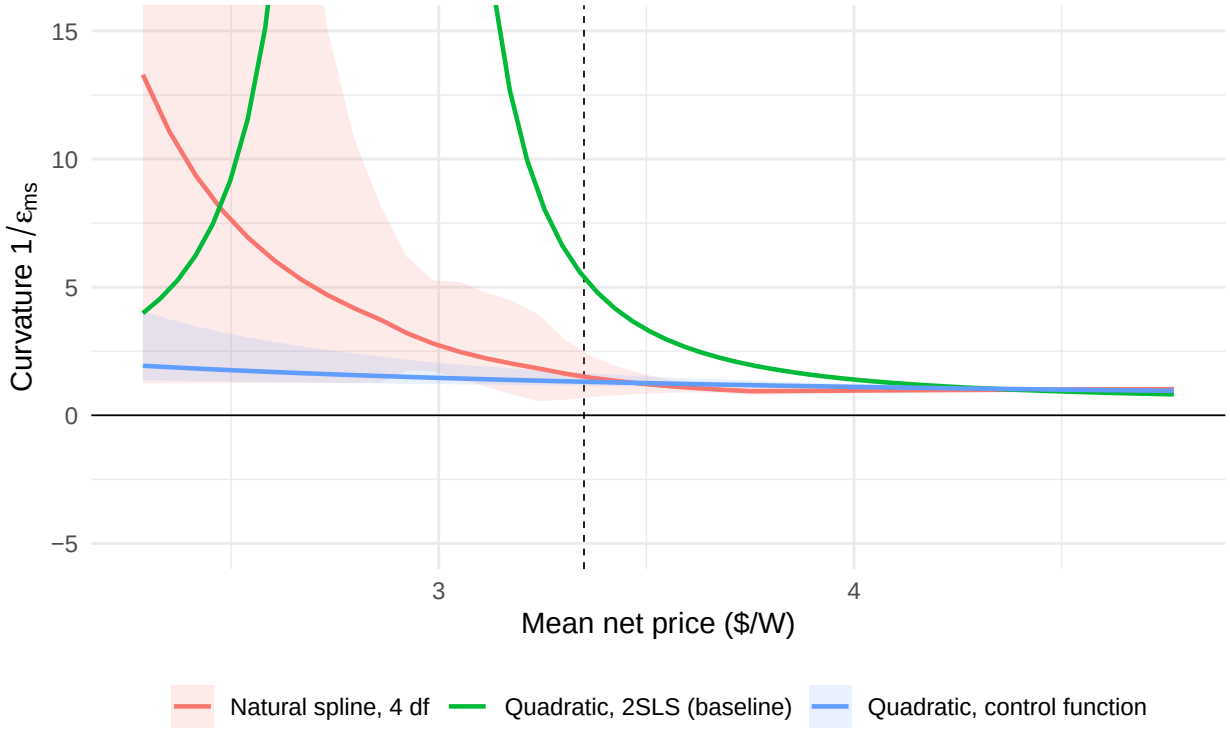



Figure 11: Curvature of SMART Demand across Prices. Curvature $1/\varepsilon_{ms}$ evaluated along each fitted demand curve between the 5th and 95th percentiles of the net price. Shaded areas are 90% bootstrap intervals for the control-function estimates. The dashed line marks the mean net price.

# E Pass-Through of a Roof-Specific Subsidy

This appendix derives the expression for roof-specific pass-through in Section 8.1. Group buyers by roof type $k$, which determines the subsidy $s_k$ a buyer receives, and let $D_k(p)$ be the probability that a buyer of type $k$ installs at net price $p = P_k - s_k$, where $P_k$ is the gross price the installer charges. The installer observes $k$, because it computes the roof's expected output itself, and its cost of installing roof $k$ is $c_k$, which does not depend on the subsidy. Following Weyl and Fabinger (2013), I summarize conduct by a parameter $\theta_B \in [0, 1]$ in the pricing condition

$$P_k - c_k = \theta_B \, m_k(P_k - s_k), \qquad m_k(p) \equiv -\frac{D_k(p)}{D_k'(p)},$$

which is marginal-cost pricing when $\theta_B = 0$ and monopoly pricing of roof $k$ when $\theta_B = 1$. Intermediate values describe competition among the installers that quote the roof. Because one roof is a negligible share of an installer's volume, a change in $s_k$ does not move the installer along its marginal cost curve, so $c_k$ is fixed.

Differentiating with respect to $s_k$,

$$\frac{dP_k}{ds_k} = \theta_B \, m_k'(p) \left( \frac{dP_k}{ds_k} - 1 \right) \quad \Longrightarrow \quad \frac{dP_k}{ds_k} = \frac{-\theta_B m_k'}{1 - \theta_B m_k'}.$$

Since $m_k' = -(1 - D_k D_k''/D_k'^2) = -1/\varepsilon^B_{ms}$, the pass-through of the roof's subsidy to the buyer's net price is

$$\rho_B = 1 - \frac{dP_k}{ds_k} = \frac{1}{1 + \theta_B/\varepsilon^B_{ms}}, \qquad \text{so} \qquad \frac{1}{\rho_B} - 1 = \frac{\theta_B}{\varepsilon^B_{ms}}.$$

For a subsidy that changes for all buyers at once, the installer's total volume changes, and the slope of marginal cost adds the term $r$ of equation (2). With $r \geq 0$ and the same conduct and curvature in the two expressions, the market-wide rate is therefore no higher than the roof-specific rate.

Three cases give $\rho_B = 1$ regardless of the installer's markup. Under perfect competition, $\theta_B = 0$. If the installer charges one price per kW for every roof, $P_k$ cannot depend on $s_k$. And

if demand is log-linear, $1/\varepsilon^B_{ms} = 0$, and the markup $m_k$ is a constant number of dollars. Pass-through below one therefore requires both pricing by roof and log-concave demand among buyers with similar roofs.

Two qualifications apply to the bound. First, if installation cost varies with the subsidy across roofs, $c_k = c + \kappa s_k$, then $dP_k/ds_k$ gains the term $\kappa/(1 - \theta_B m'_k)$, and the estimated shortfall $1 - \hat{\rho}_B$ overstates the shortfall due to conduct when $\kappa > 0$ and understates it when $\kappa < 0$. Section 7.1 shows that, holding tilt fixed, cost differences across roofs leave the SMART estimate unchanged. Second, the bound uses the curvature of aggregate ZIP-level demand in place of $\varepsilon^B_{ms}$, the curvature among buyers with similar roofs, which the data do not identify separately.

# F Price Discrimination Across Roofs

Installers quote each project individually and compute the roof's expected output themselves, as the production estimates on SMART applications show, so the installer knows the subsidy a buyer will receive. Section 8.1 shows that the cross-roof estimates measure the pass-through of a roof-specific subsidy, which is below one only if the gross price rises with the roof's subsidy, and Table 10 shows that installers price the roof only a little. Under SMART the rate is also known with certainty at the time of sale. If certainty made pricing by roof easier, cross-roof pass-through would be lower under SMART than under Solar Carve-Out. The two estimates are nearly identical (0.85 and 0.86), which gives no indication that it did.

## F.1 Welfare Effects of Pricing by Roof

If installers do price by roof, a natural question is whether buyers as a group gain or lose relative to a uniform price. Aguirre, Cowan and Vickers (2010) and Miklós-Thal and Shaffer (2021) show that the answer depends on pass-through and on the curvature of demand in each market. Treat roofs with high generation as the strong market, where the discriminatory price is higher, and roofs with low generation as the weak market. Relative to uniform pricing, discrimination raises output if pass-through is higher in the weak market ($\rho_W > \rho_S$, averaged over the relevant range) and inverse demand is convex in both markets. It lowers output if $\rho_W < \rho_S$ and inverse demand is concave in both markets. The same conditions, with pass-through weighted by the markup, determine the welfare effect. Miklós-Thal and Shaffer (2021) show that the ranking of pass-through rates alone is not sufficient, so the sign of curvature is needed. Demand under SMART is concave (Table 27). If pass-through were lower on roofs with low generation, pricing by roof would therefore reduce adoption relative to a uniform price per kW. The test in Appendix B.7 finds no such difference, so the data do not sign this effect. Miklós-Thal and Shaffer (2021) also note that formulas based on

pass-through hold exactly only for small changes unless demand is linear, so the division of surplus in Section 8.2, which is evaluated at the mean, is an approximation for a subsidy of this size.